FEMB QC Criteria Report

FEMB-IO 1826-1L Quality Control Analysis

Analysis Date: October 2026 **Board Version:** FEMB-IO 1826-1L **Number of Boards Analyzed:** 44 **Total Parameters:** 44 **Analysis Method:** Statistical analysis using latest passing test per board

Overview

This report documents the Quality Control (QC) criteria derived from 44 FEMB-IO 1826-1L boards that passed checkout testing at BNL.

Tolerance Methods

Different tolerance methods are used based on parameter type:

Parameter Type	Tolerance Method	Rationale
128-Channel (ped, rms, etc.)	5σ	Statistical outlier detection for large datasets (5632 points)
8-Chip ASIC	5σ	Statistical outlier detection (352 points)
LDO Voltages	7σ	Wider tolerance for power rail monitoring (44 points)
Currents	7σ	Wider tolerance for current measurements (44 points)
Power	7σ	Wider tolerance for power measurements (44 points)

Parameter Categories

Category	Description	Count
128-Channel	Per-channel measurements (128 ch × 44 boards = 5632 points)	7
LDO Pedestal	LDO voltage readings during pedestal check	11
LDO Pulse	LDO voltage readings during pulse check	11
Current	Power rail current measurements	6
Power	Total power measurements	2
8-Chip ASIC	Per-ASIC measurements (8 chips × 44 boards = 352 points)	7

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1. 128-Channel Parameters (5σ Tolerance)

These parameters are measured for all 128 channels on each FEMB board (5632 total data points).

1.1 Pedestal (ped)

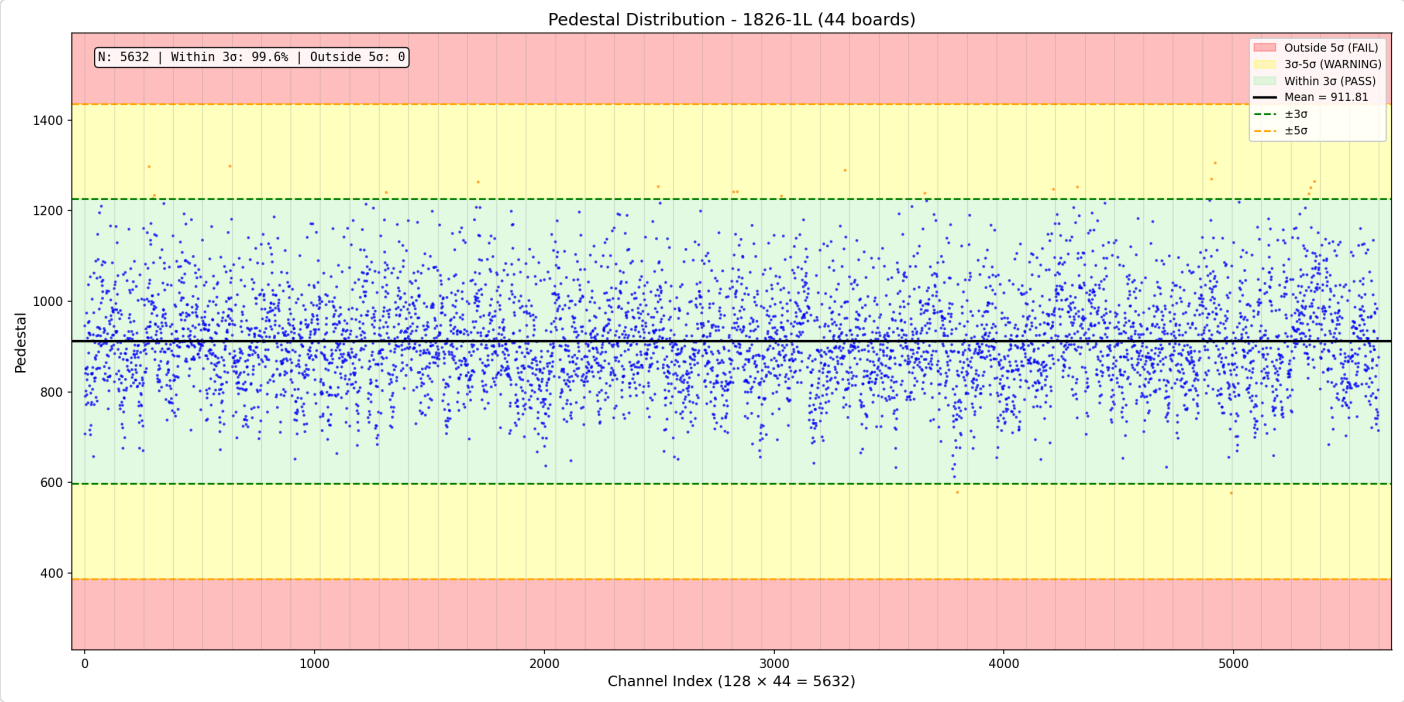
Description: The baseline ADC value when no signal is present. Represents the DC offset of each channel's readout electronics.

Statistic	Value
Mean	911.8066
Std	104.7993
Median	903.3195
Min	576.8393
Max	1305.4897
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
387.8101	1435.8031

Interpretation: Pedestal values vary across channels due to manufacturing variations.



1.2 RMS Noise (rms)

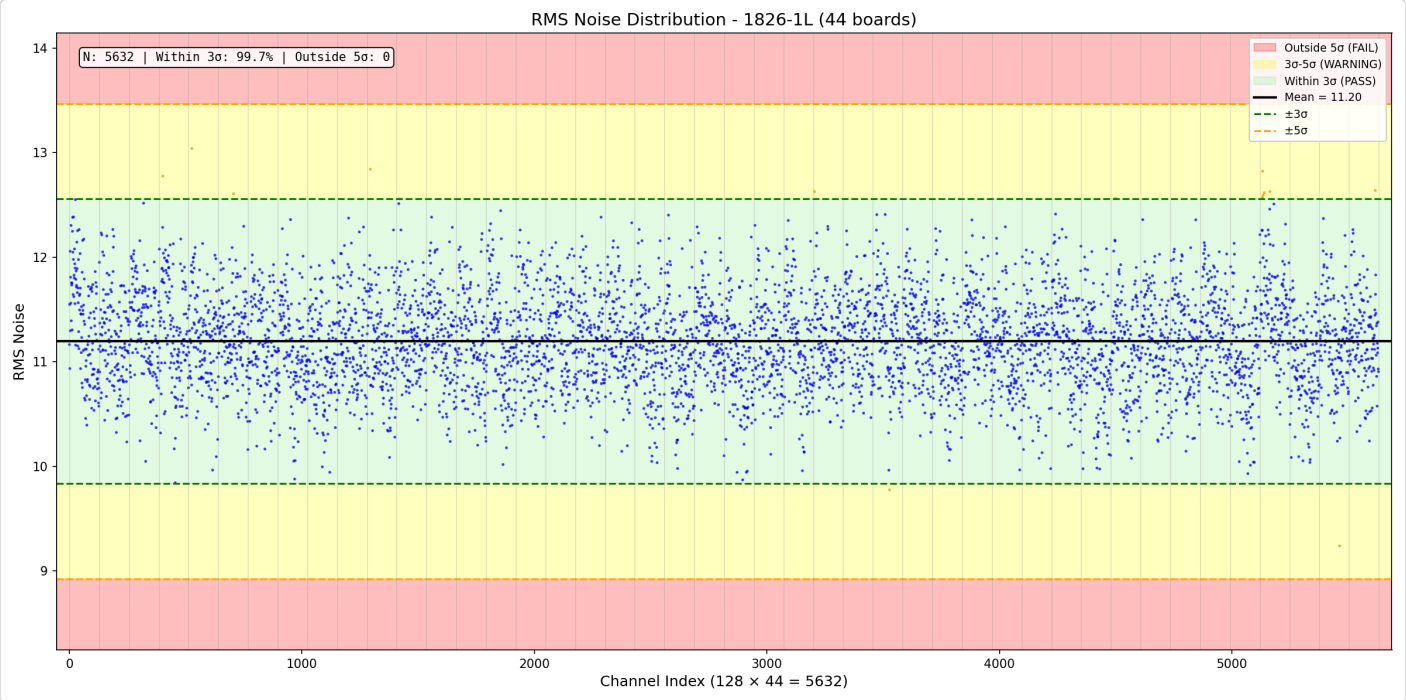
Description: Root Mean Square of the baseline fluctuations. Measures the intrinsic electronic noise of each channel in ADC counts.

Statistic	Value
Mean	11.1965
Std	0.4542
Median	11.1850
Min	9.2394
Max	13.0419
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
8.9255	13.4675

Interpretation: Lower noise is better. High RMS indicates noisy channels.



1.3 Positive Peak (PPK_mean)

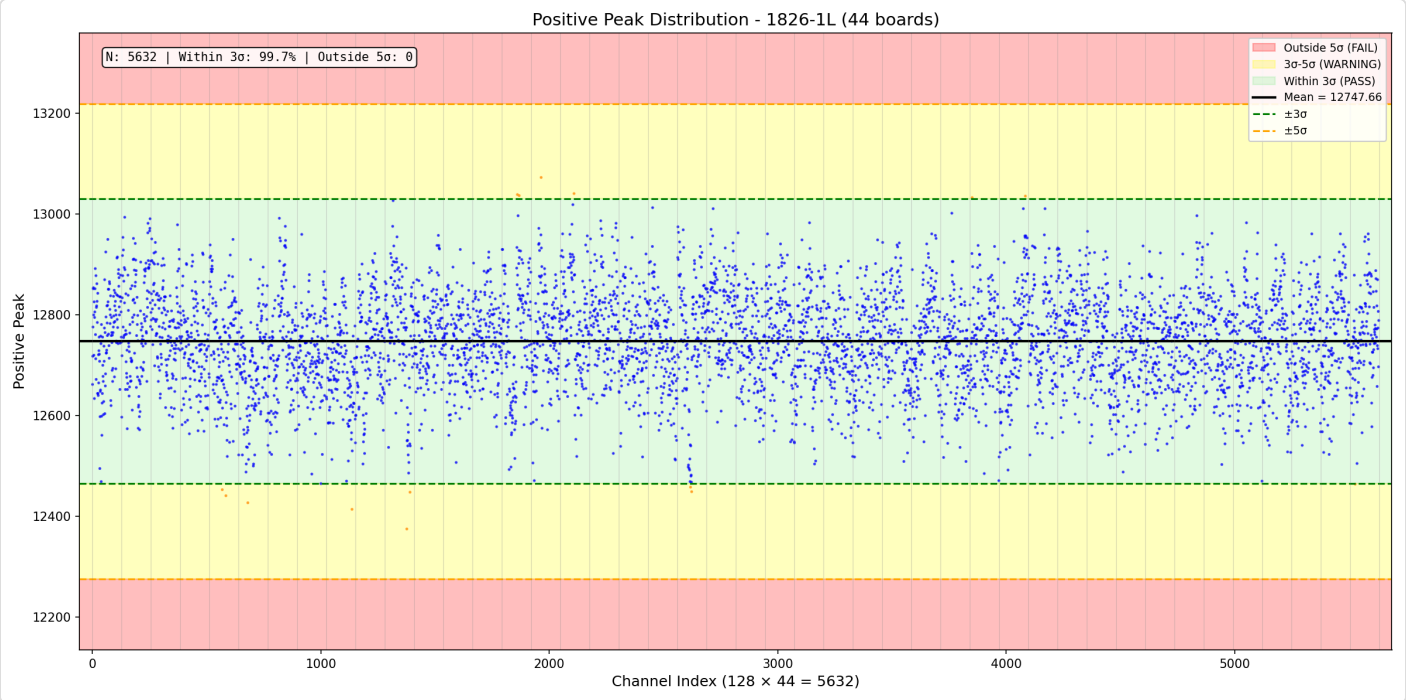
Description: Mean ADC value of the positive pulse response when injecting a calibration pulse.

Statistic	Value
Mean	12747.6571
Std	94.2848
Median	12749.0000
Min	12375.0000
Max	13073.0000
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
12276.2331	13219.0811

Interpretation: Consistent PPK values indicate uniform gain.



1.4 Negative Peak (NPK_mean)

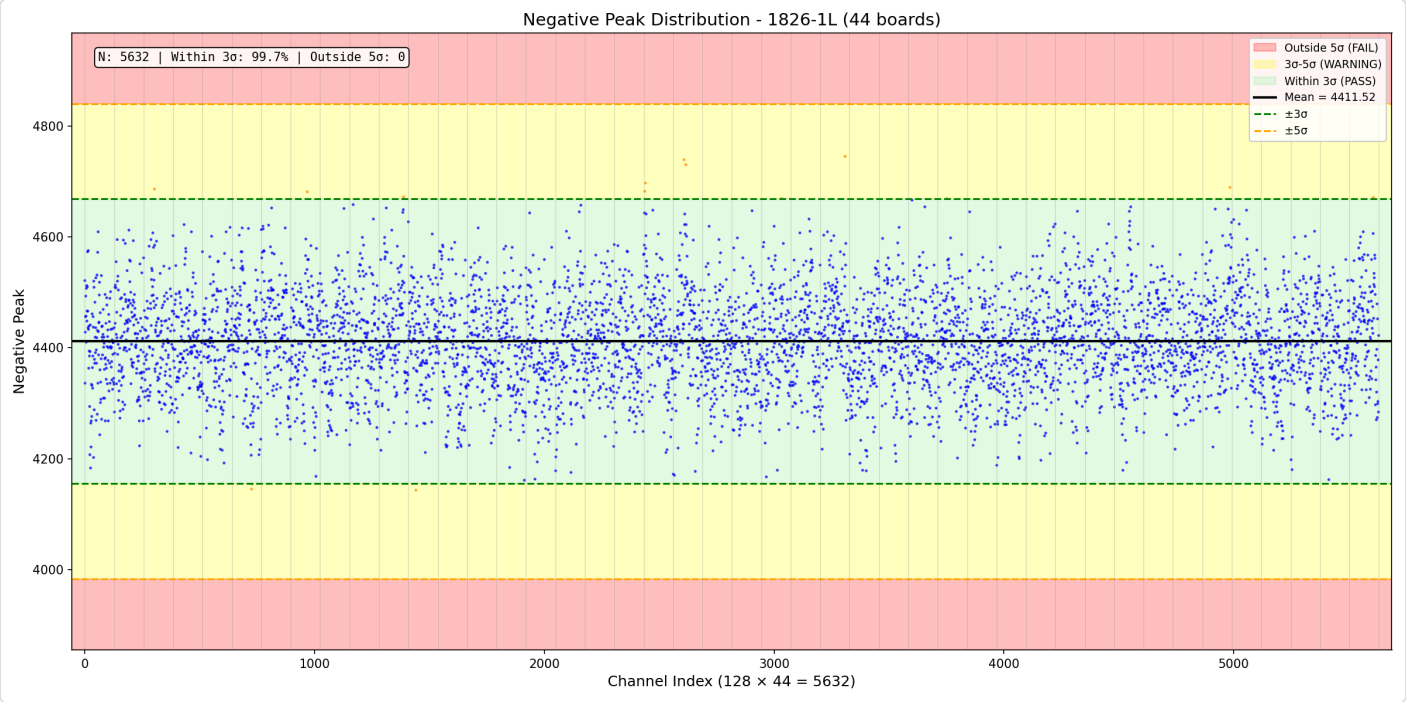
Description: Mean ADC value of the negative pulse response.

Statistic	Value
Mean	4411.5245
Std	85.6226
Median	4411.0000
Min	4143.0000
Max	4745.0000
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
3983.4115	4839.6375

Interpretation: NPK should be symmetric with PPK relative to baseline.



1.5 Baseline (BL_mean)

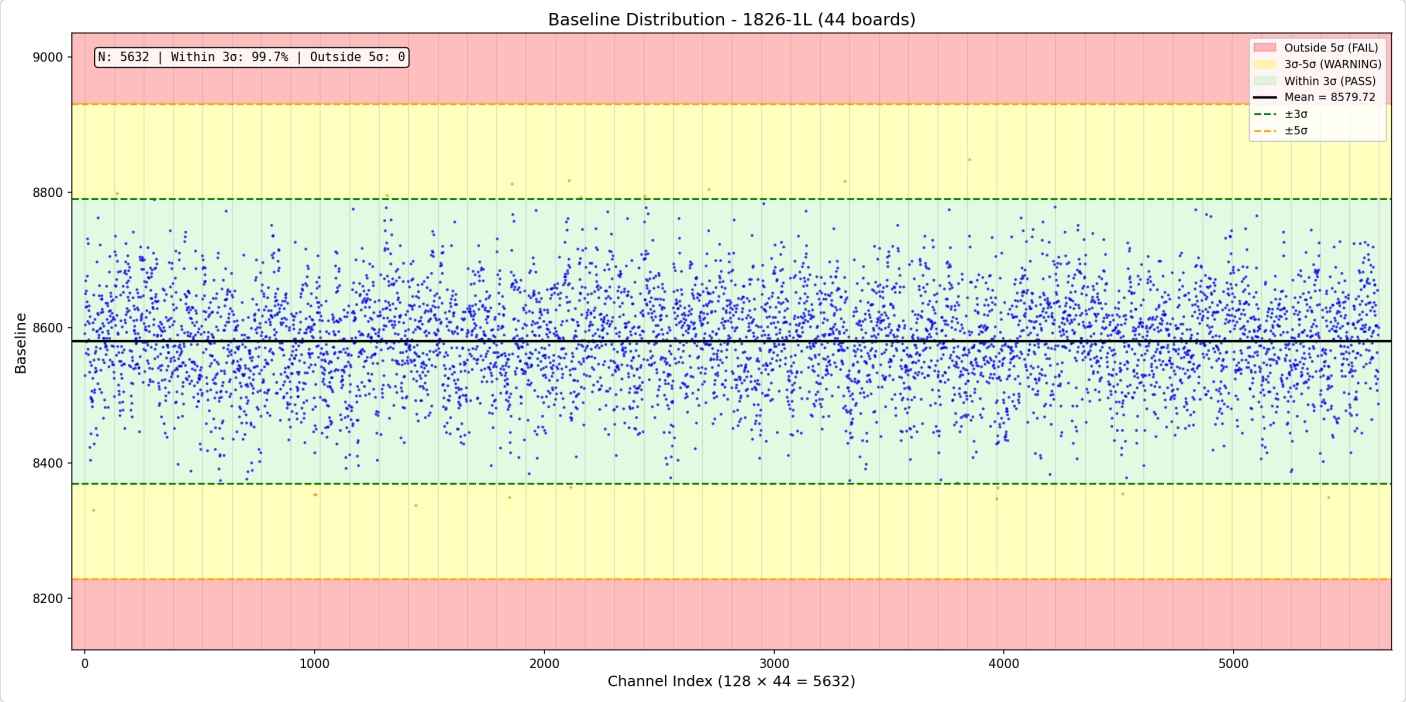
Description: Mean baseline level during pulse response measurement.

Statistic	Value
Mean	8579.7246
Std	70.1446
Median	8581.0000
Min	8330.0000
Max	8848.0000
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
8229.0016	8930.4476

Interpretation: Stable baseline indicates good DC performance.



1.6 Pedestal Max (pedmax)

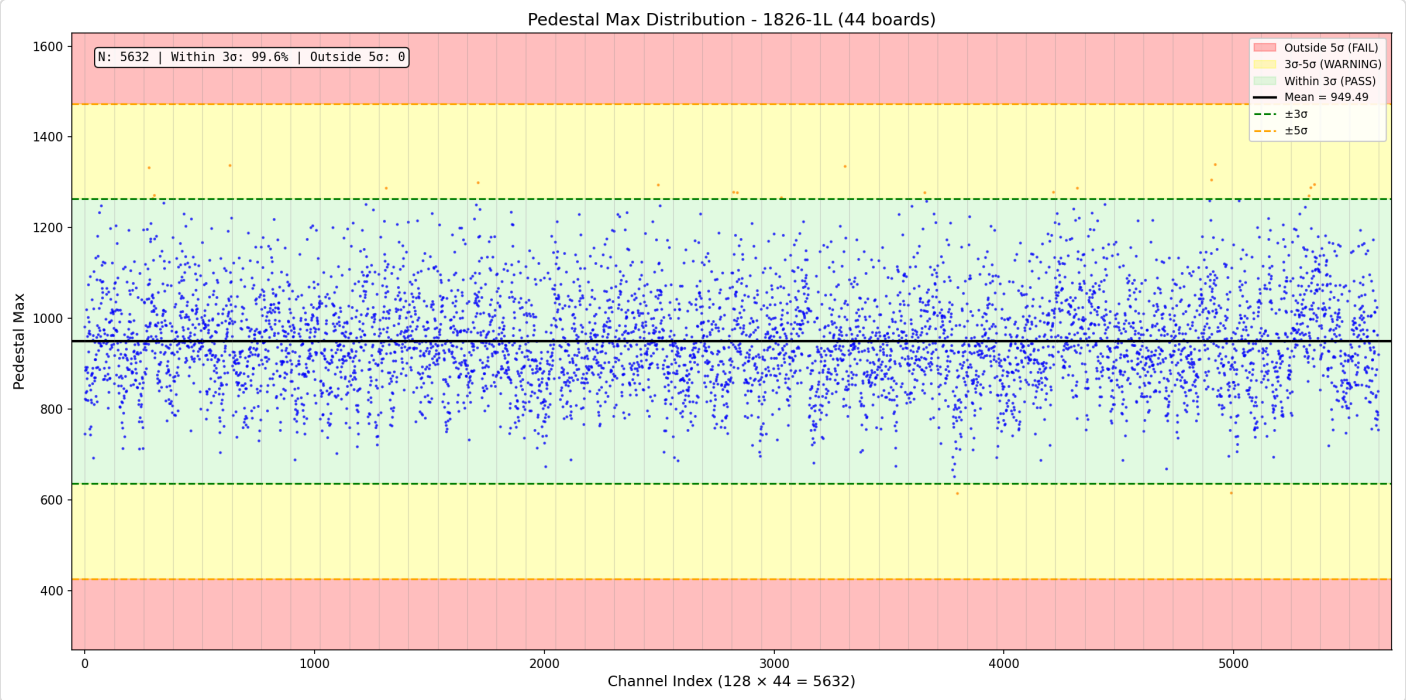
Description: Maximum pedestal value observed during the measurement window.

Statistic	Value
Mean	949.4876
Std	104.6585
Median	941.0000
Min	614.0000
Max	1339.0000
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
426.1951	1472.7801

Interpretation: Large pedmax-pedmin difference indicates instability.



1.7 Pedestal Min (pedmin)

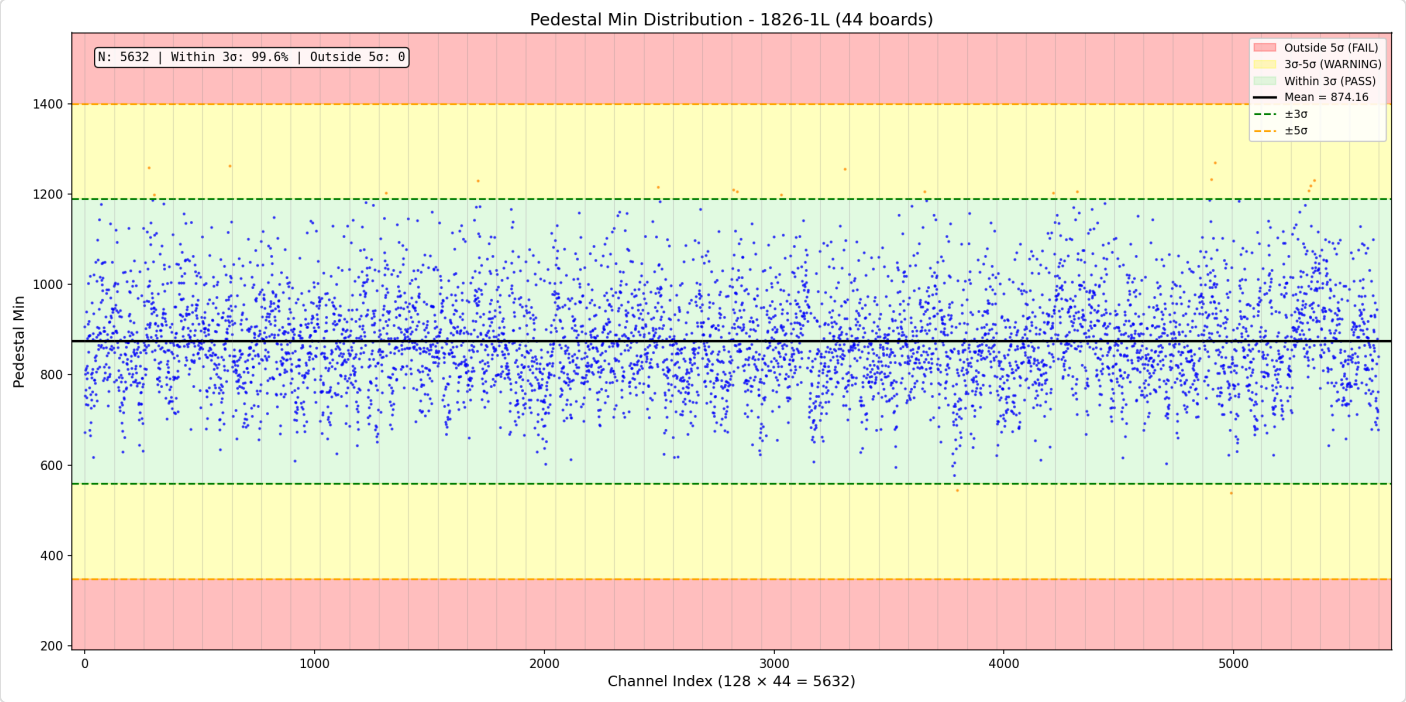
Description: Minimum pedestal value observed during the measurement window.

Statistic	Value
Mean	874.1628
Std	105.0868
Median	865.0000
Min	538.0000
Max	1269.0000
N	5632

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
348.7288	1399.5968

Interpretation: Together with pedmax, defines pedestal variation range.



2. LDO Voltage Parameters - Pedestal (7σ Tolerance)

LDO regulator output voltages measured during pedestal check.

2.1 CDVDDA_ped (CDVDDA_ped)

Description: COLDATA analog supply voltage (VDDA).

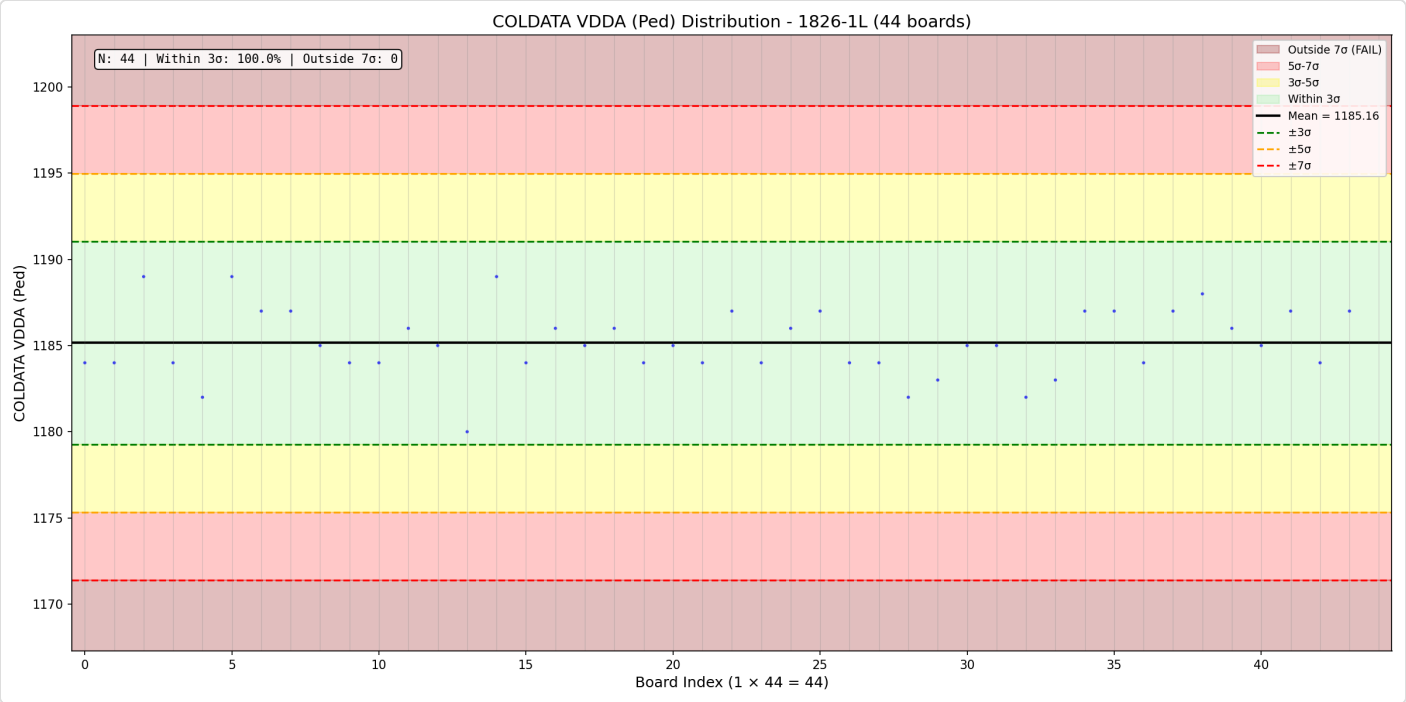
Statistic	Value
Mean	1185.1591
Std	1.9650
Median	1185.0000
Min	1180.0000

Statistic	Value
Max	1189.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1171.4041	1198.9141

Interpretation: Should be stable within 7σ bounds.



2.2 CDVDDIO_HALF_ped (CDVDDIO_HALF_ped)

Description: COLDATA I/O supply voltage (divided by 2).

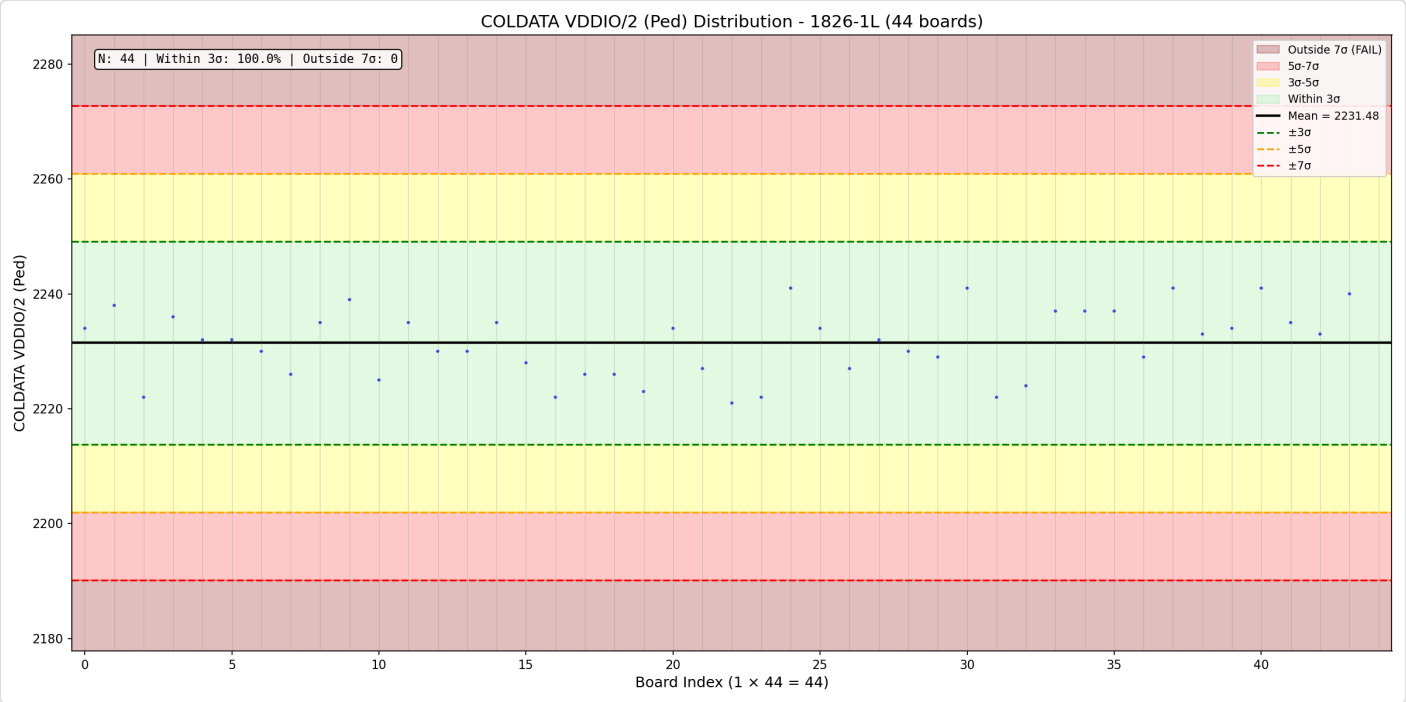
Statistic	Value
Mean	2231.4773
Std	5.8949
Median	2232.0000

Statistic	Value
Min	2221.0000
Max	2241.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2190.2130	2272.7416

Interpretation: Should be stable within 7σ bounds.



2.3 ADCRVDDD1P2_ped (ADCRVDDD1P2_ped)

Description: Right ColdADC digital 1.2V supply.

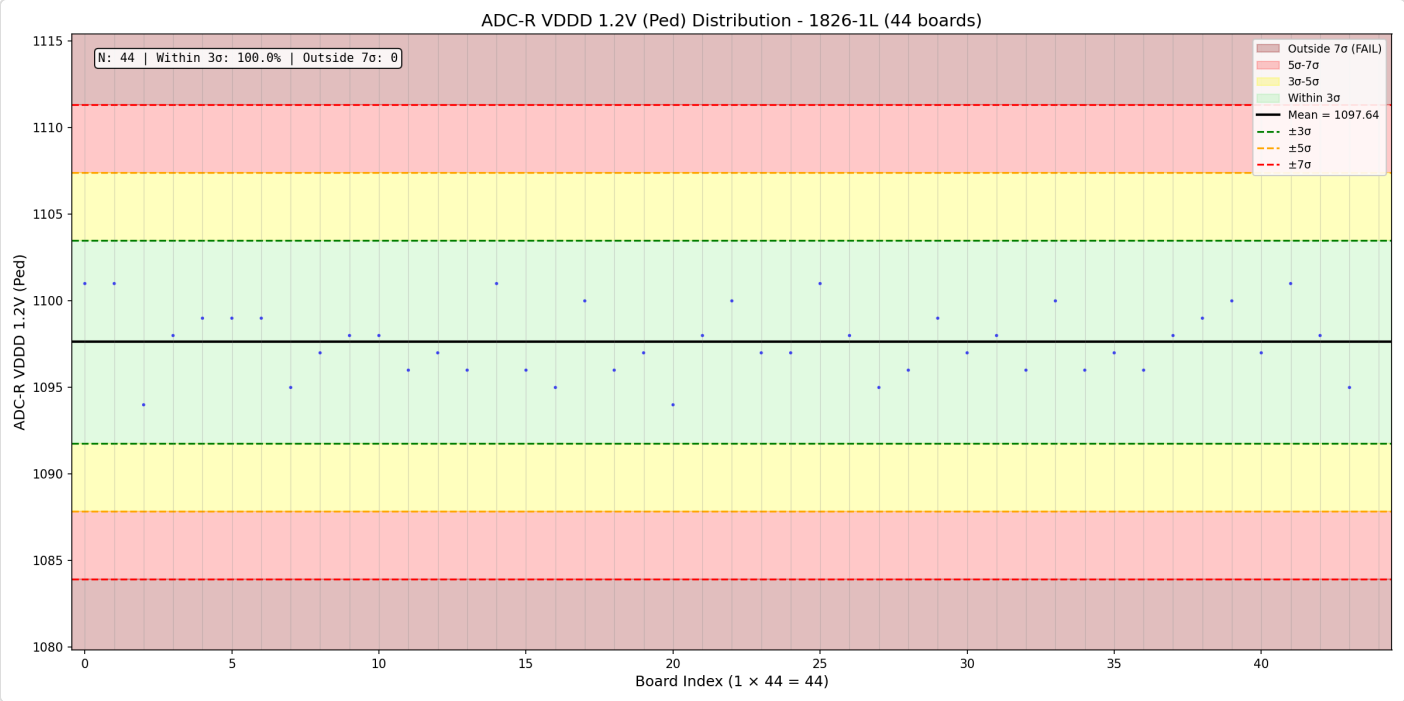
Statistic	Value
Mean	1097.6364
Std	1.9551

Statistic	Value
Median	1097.5000
Min	1094.0000
Max	1101.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1083.9507	1111.3221

Interpretation: Should be stable within 7σ bounds.



2.4 ADCLVDDD1P2_ped (ADCLVDDD1P2_ped)

Description: Left ColdADC digital 1.2V supply.

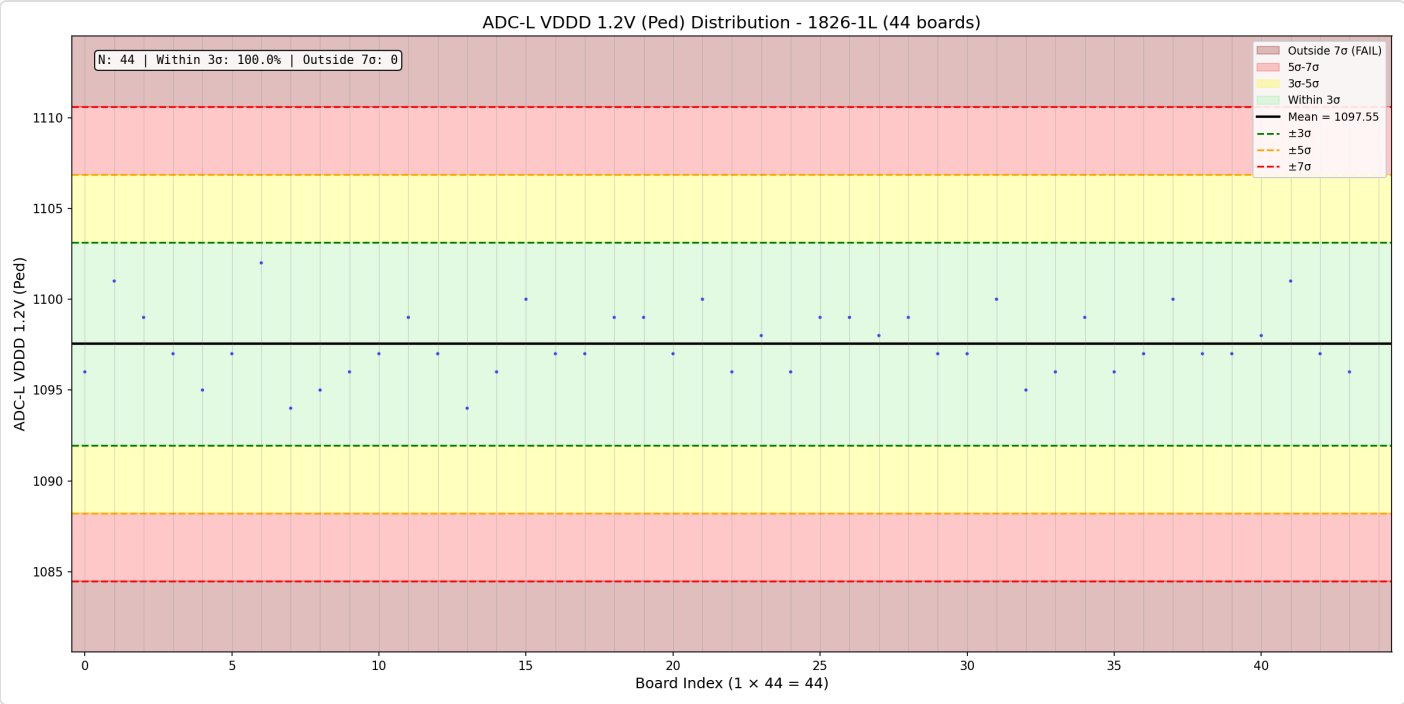
Statistic	Value
Mean	1097.5455

Statistic	Value
Std	1.8642
Median	1097.0000
Min	1094.0000
Max	1102.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1084.4961	1110.5949

Interpretation: Should be stable within 7σ bounds.



2.5 FERVDDP_ped (FERVDDP_ped)

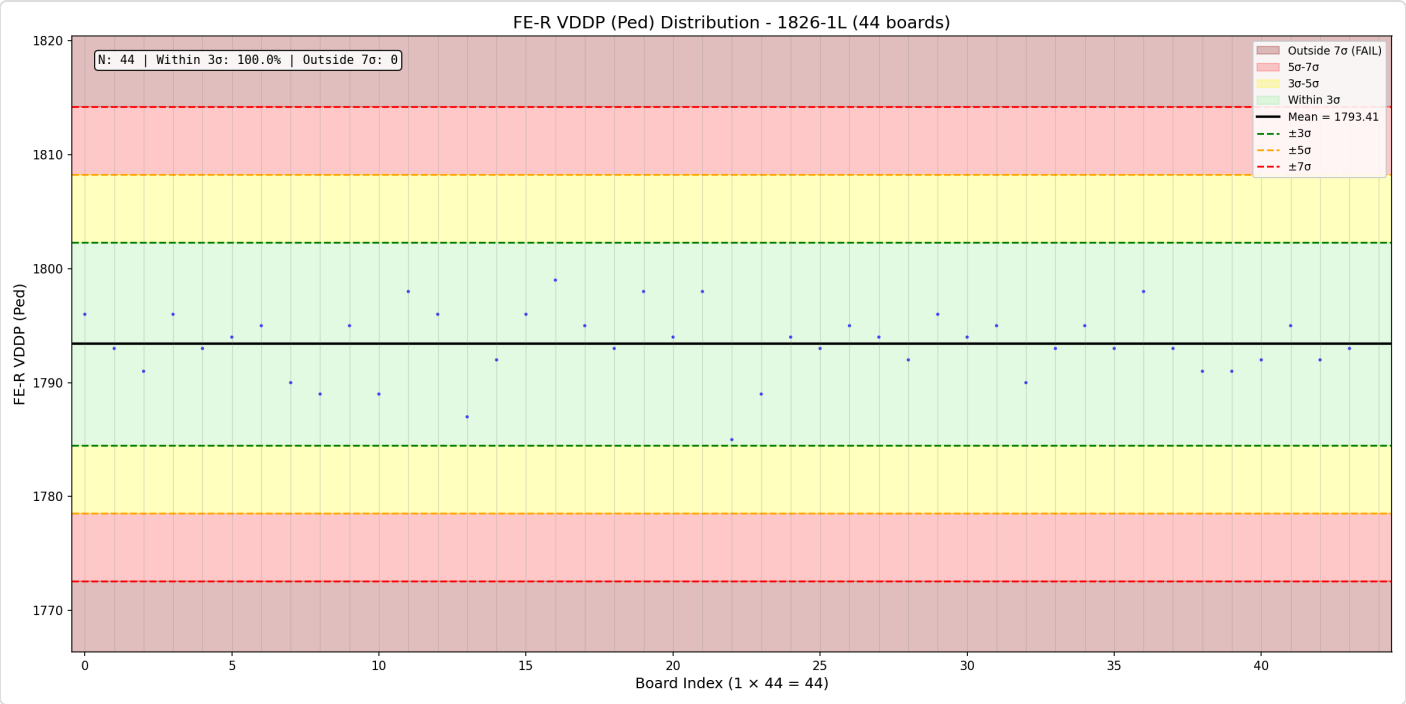
Description: Right LArASIC positive supply voltage.

Statistic	Value
Mean	1793.4091
Std	2.9720
Median	1793.5000
Min	1785.0000
Max	1799.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1772.6051	1814.2131

Interpretation: Higher during pedestal, lower during pulse.



2.6 FELVDDP_ped (FELVDDP_ped)

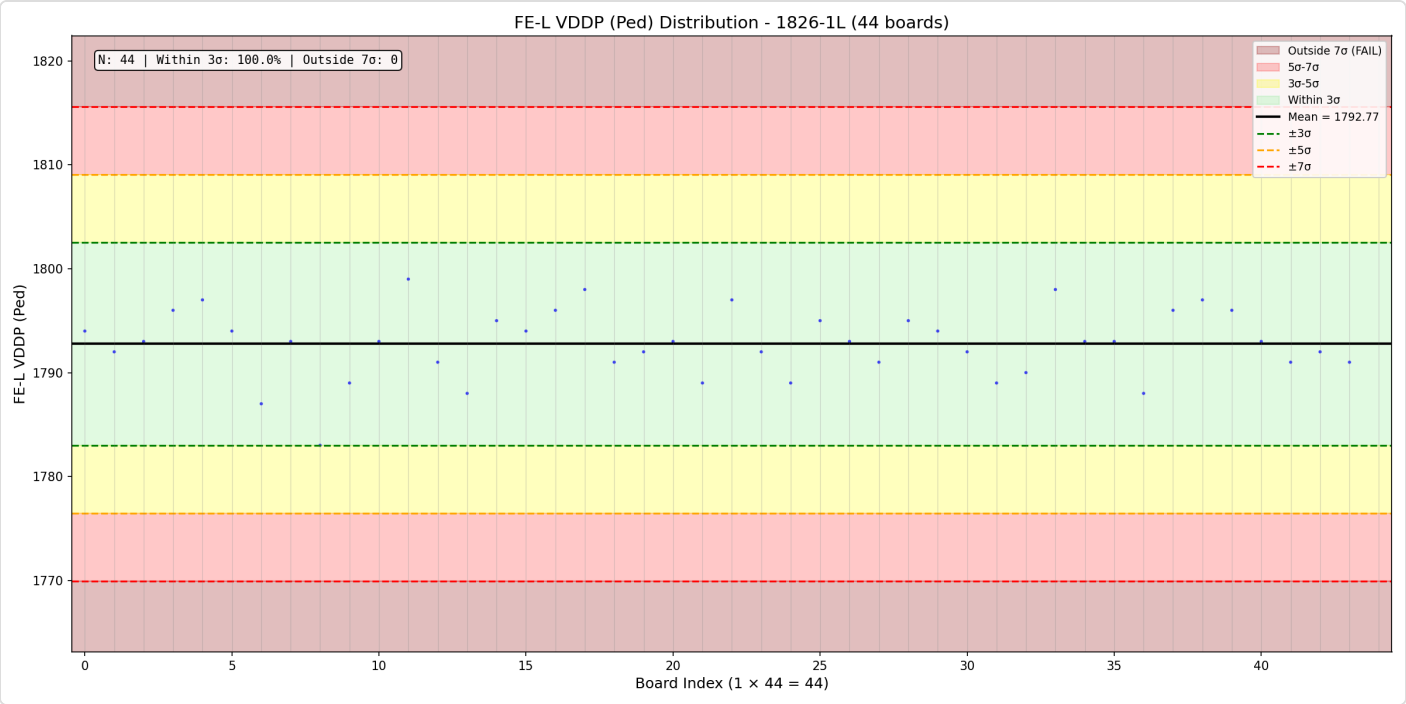
Description: Left LArASIC positive supply voltage.

Statistic	Value
Mean	1792.7727
Std	3.2604
Median	1793.0000
Min	1783.0000
Max	1799.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1769.9499	1815.5955

Interpretation: Higher during pedestal, lower during pulse.



2.7 ADCRP25V_1_HALF_ped (ADCRP25V_1_HALF_ped)

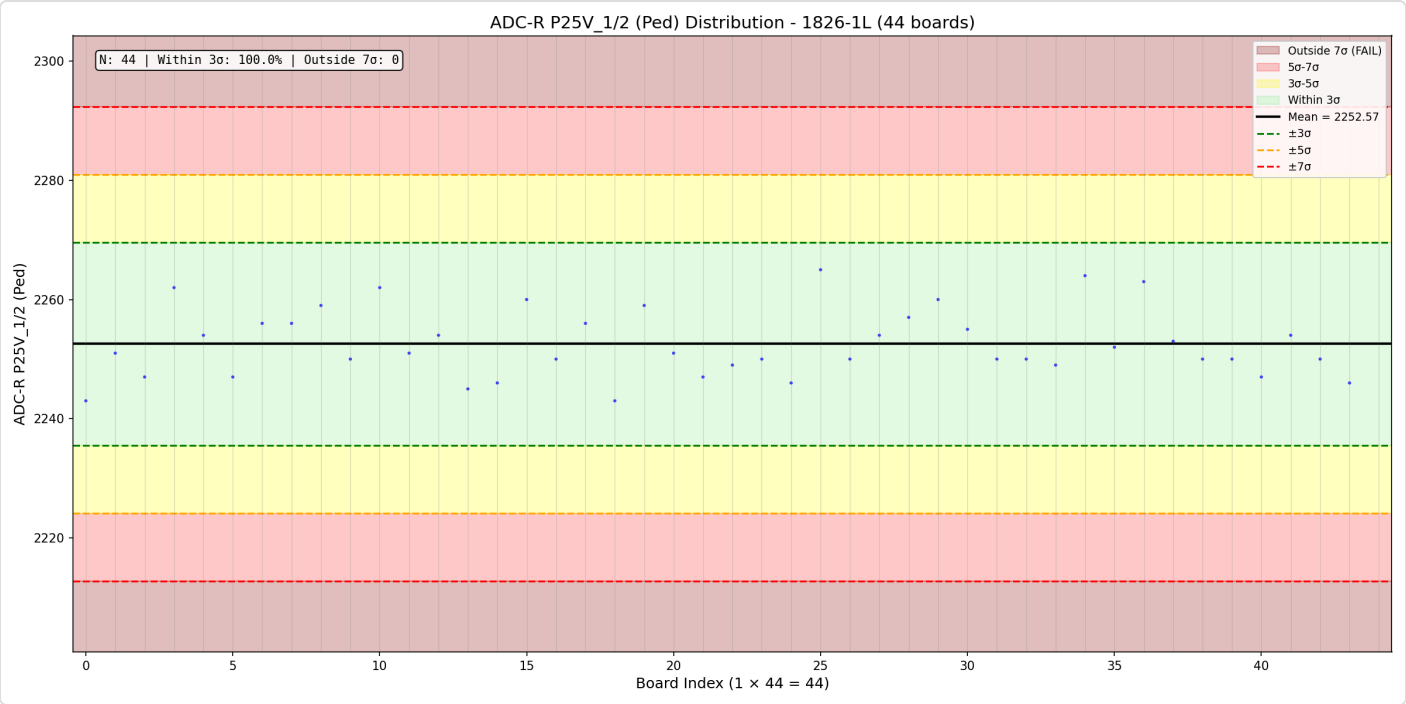
Description: Right ColdADC 2.5V supply #1 (divided by 2).

Statistic	Value
Mean	2252.5682
Std	5.6825
Median	2251.0000
Min	2243.0000
Max	2265.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2212.7907	2292.3457

Interpretation: Should be stable within 7σ bounds.



2.8 ADCLP25V_1_HALF_ped (ADCLP25V_1_HALF_ped)

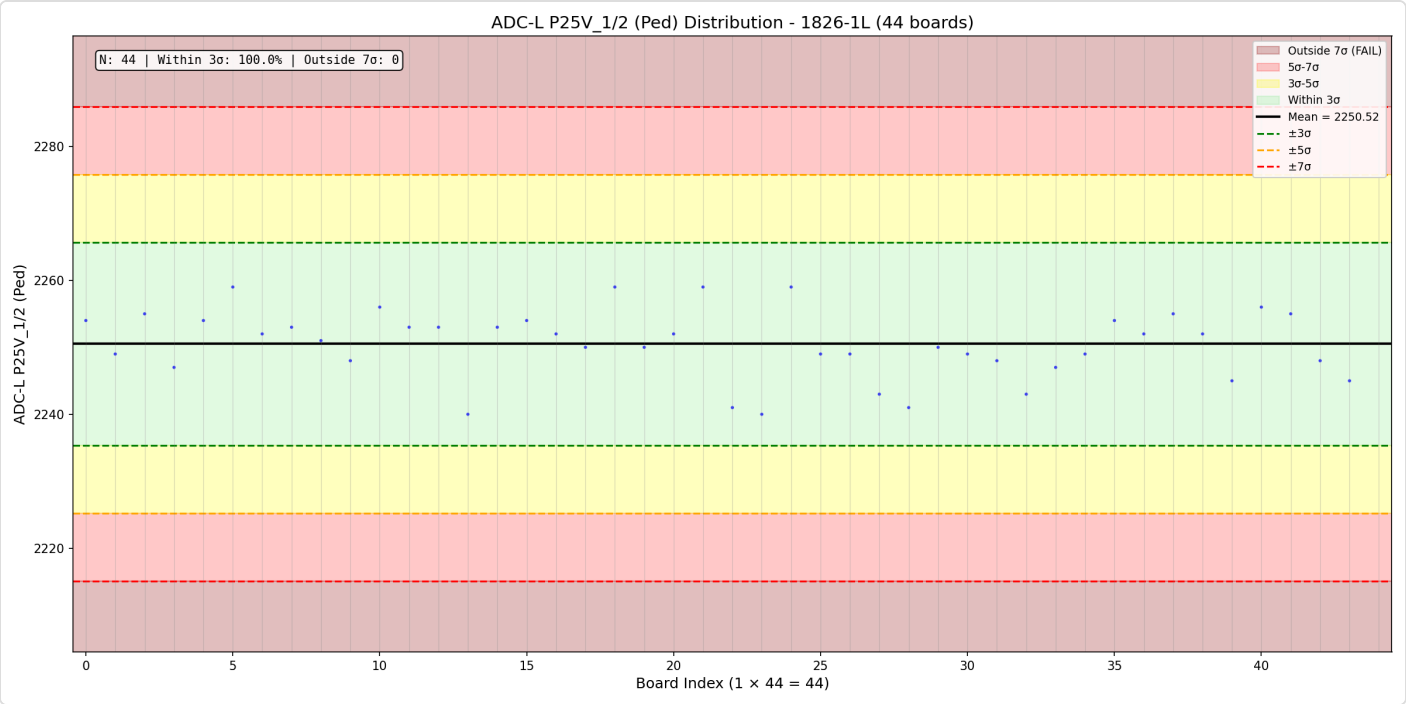
Description: Left ColdADC 2.5V supply #1 (divided by 2).

Statistic	Value
Mean	2250.5227
Std	5.0564
Median	2251.5000
Min	2240.0000
Max	2259.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2215.1279	2285.9175

Interpretation: Should be stable within 7σ bounds.



2.9 ADCRP25V_HALF_ped (ADCRP25V_HALF_ped)

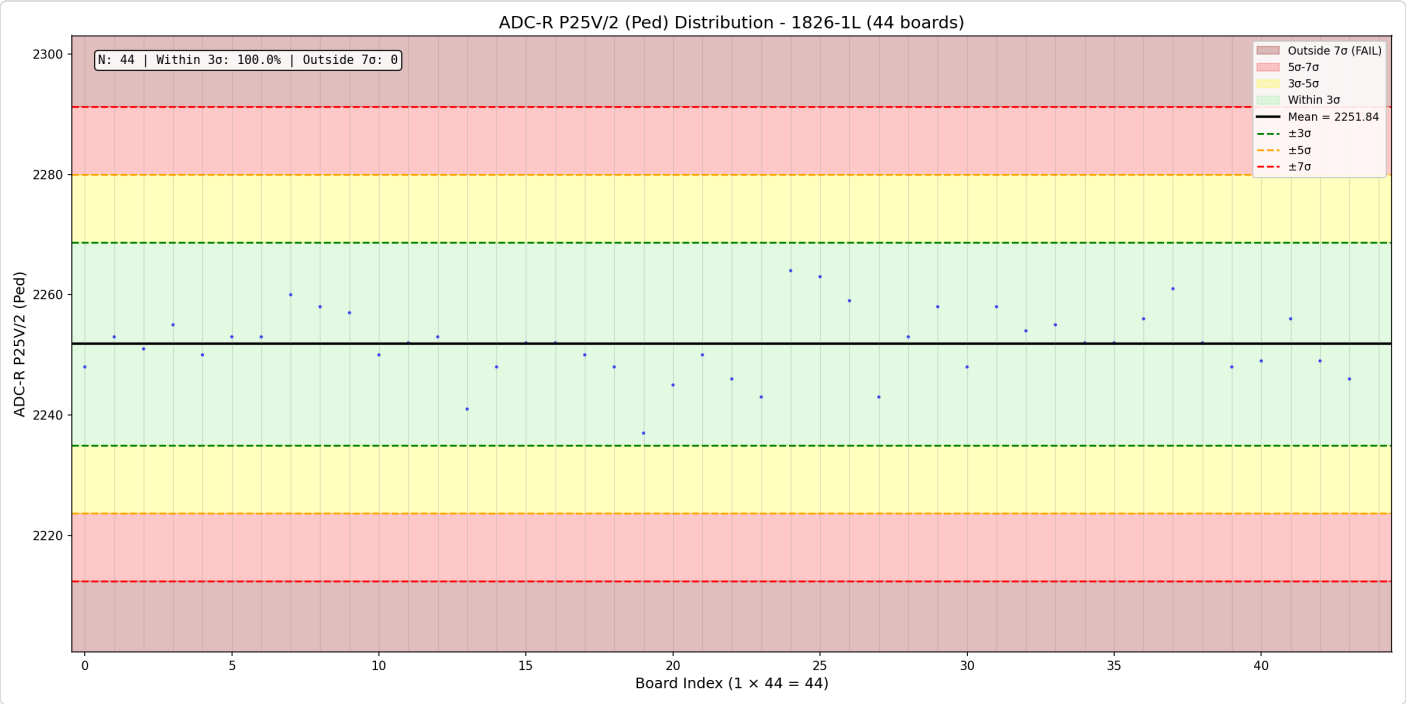
Description: Right ColdADC 2.5V main supply (divided by 2).

Statistic	Value
Mean	2251.8409
Std	5.6284
Median	2252.0000
Min	2237.0000
Max	2264.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2212.4421	2291.2397

Interpretation: Should be stable within 7σ bounds.



2.10 ADCLP25V_HALF_ped (ADCLP25V_HALF_ped)

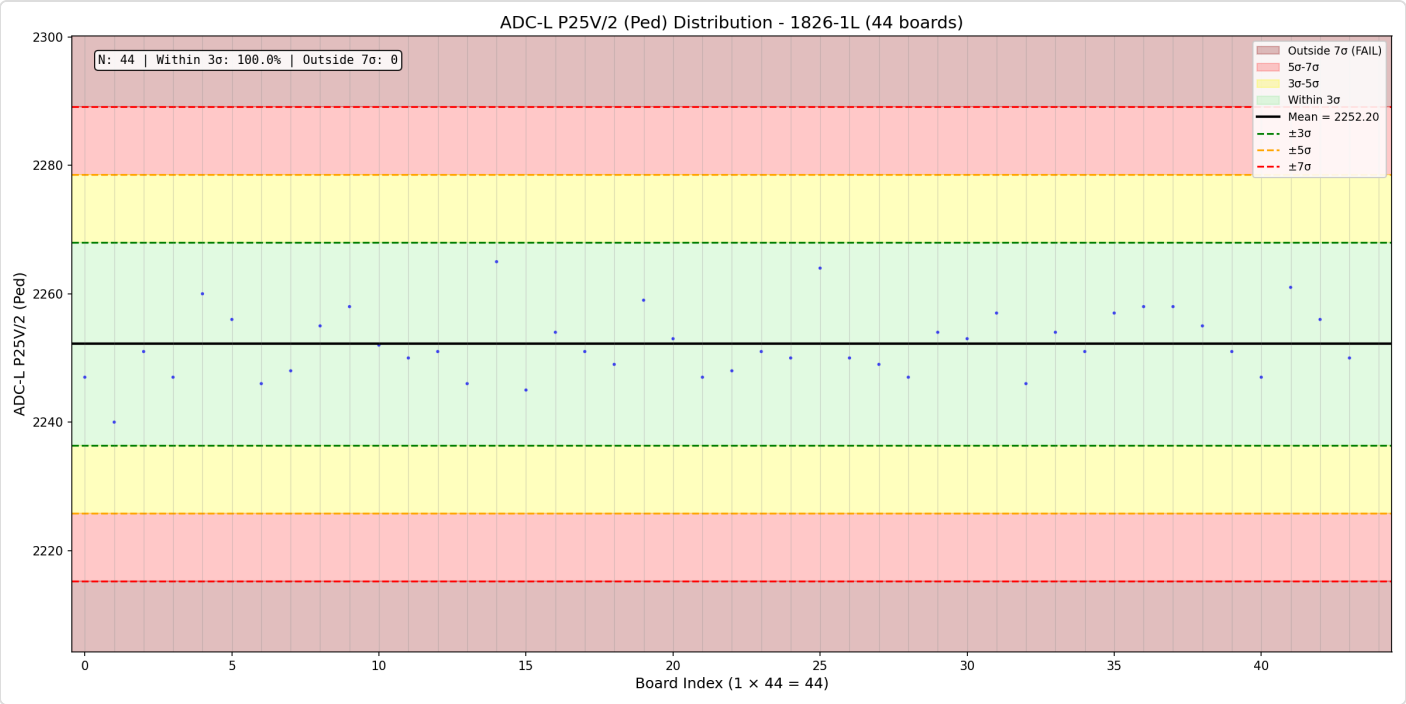
Description: Left ColdADC 2.5V main supply (divided by 2).

Statistic	Value
Mean	2252.2045
Std	5.2768
Median	2251.0000
Min	2240.0000
Max	2265.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2215.2669	2289.1421

Interpretation: Should be stable within 7σ bounds.



2.11 GND_ped (GND_ped)

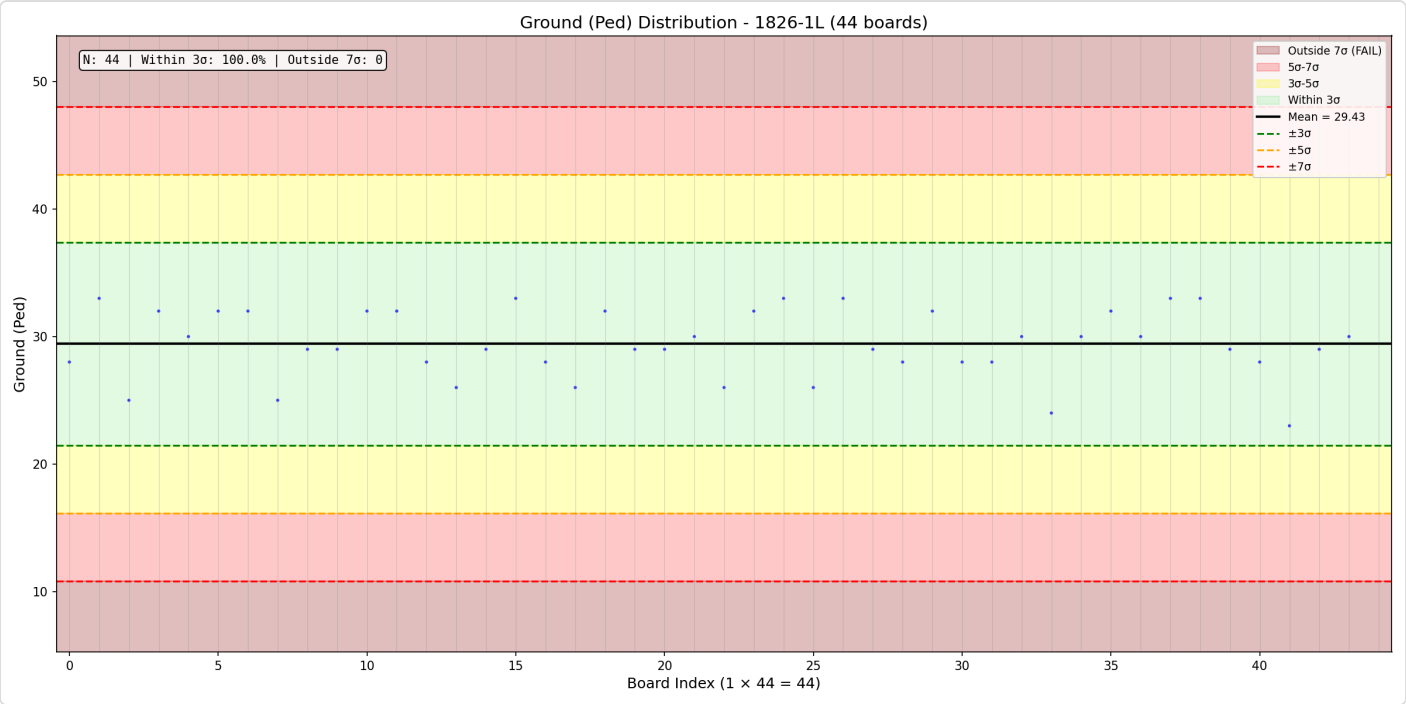
Description: Ground reference measurement.

Statistic	Value
Mean	29.4318
Std	2.6577
Median	29.0000
Min	23.0000
Max	33.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
10.8279	48.0357

Interpretation: Should be near zero, within 7σ bounds.



3. LDO Voltage Parameters - Pulse (7σ Tolerance)

LDO measurements during pulse response check.

3.1 CDVDDA_pulse (CDVDDA_pulse)

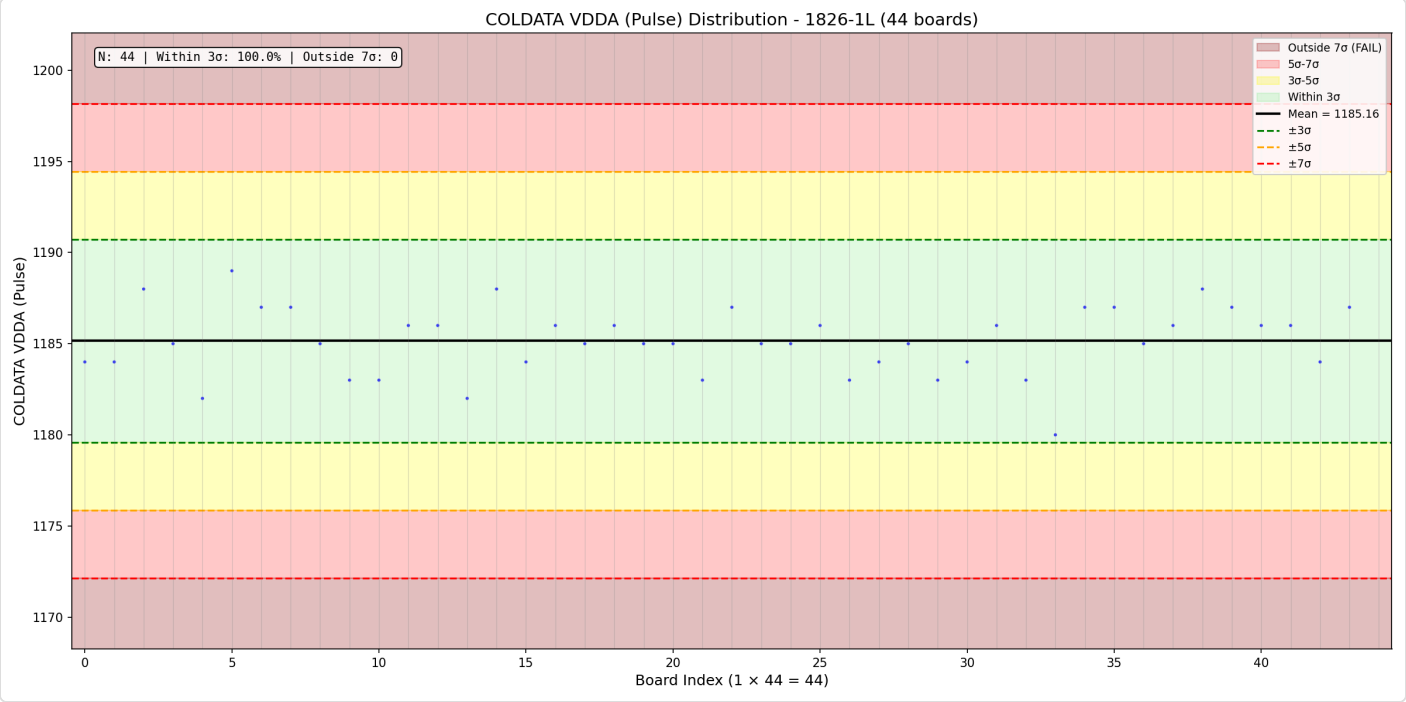
Description: COLDATA analog supply voltage (VDDA).

Statistic	Value
Mean	1185.1591
Std	1.8579
Median	1185.0000
Min	1180.0000
Max	1189.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1172.1538	1198.1644

Interpretation: Should be stable within 7σ bounds.



3.2 CDVDDIO_HALF_pulse (CDVDDIO_HALF_pulse)

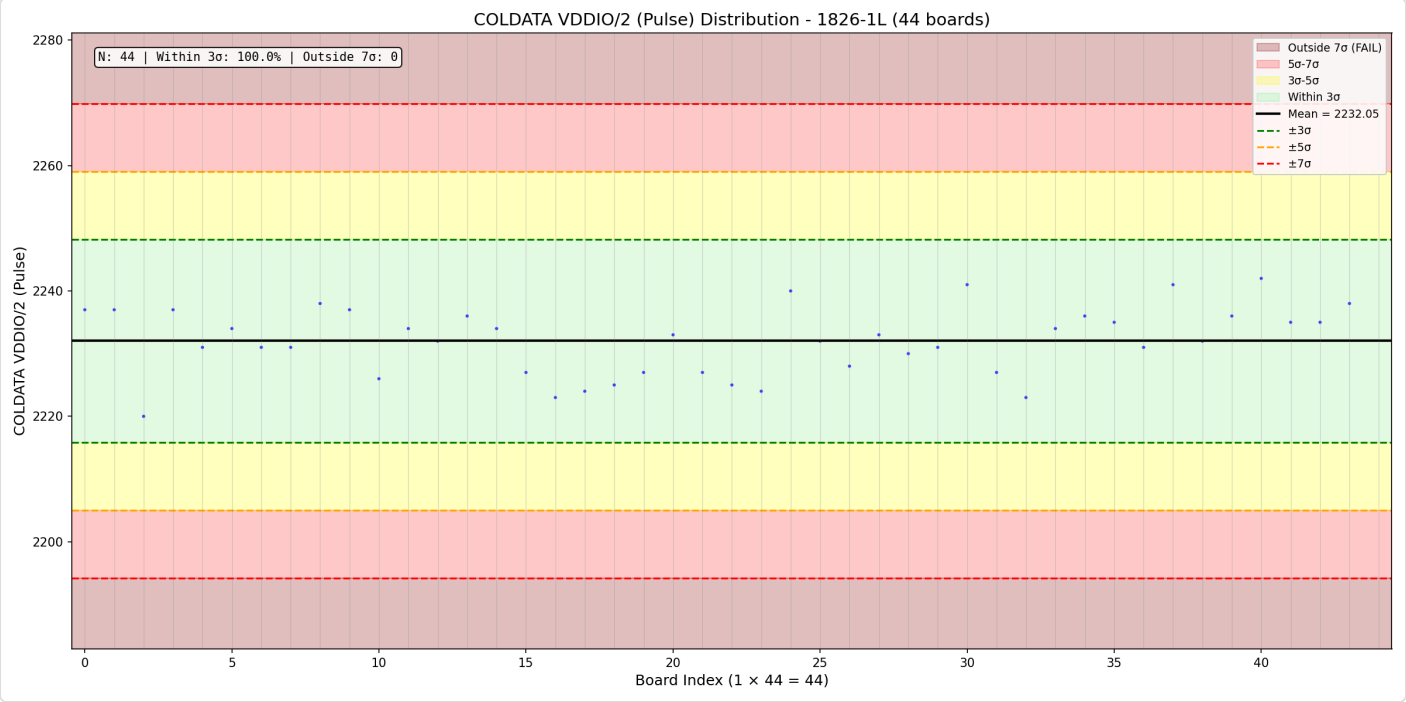
Description: COLDATA I/O supply voltage (divided by 2).

Statistic	Value
Mean	2232.0455
Std	5.4018
Median	2232.5000
Min	2220.0000
Max	2242.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2194.2329	2269.8581

Interpretation: Should be stable within 7σ bounds.



3.3 ADCRVDDD1P2_pulse (ADCRVDDD1P2_pulse)

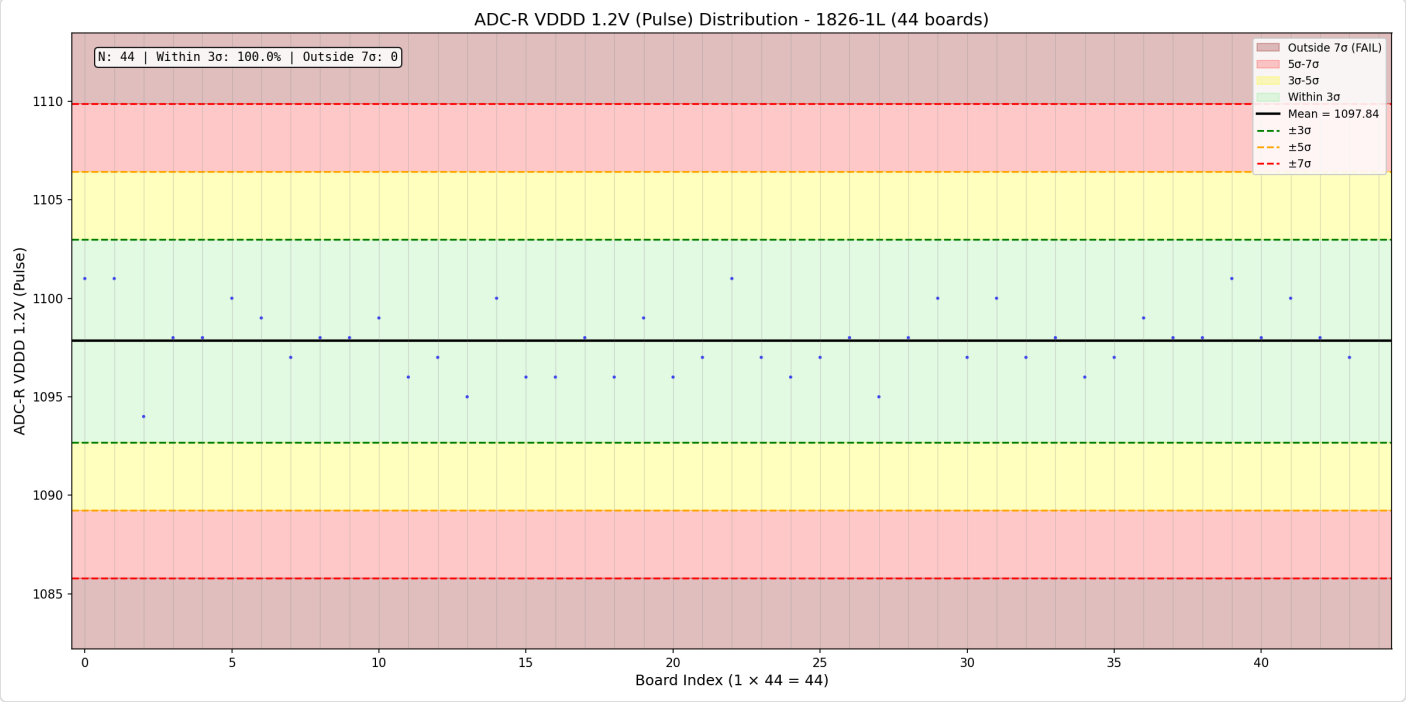
Description: Right ColdADC digital 1.2V supply.

Statistic	Value
Mean	1097.8409
Std	1.7181
Median	1098.0000
Min	1094.0000
Max	1101.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1085.8142	1109.8676

Interpretation: Should be stable within 7σ bounds.



3.4 ADCLVDDD1P2_pulse (ADCLVDDD1P2_pulse)

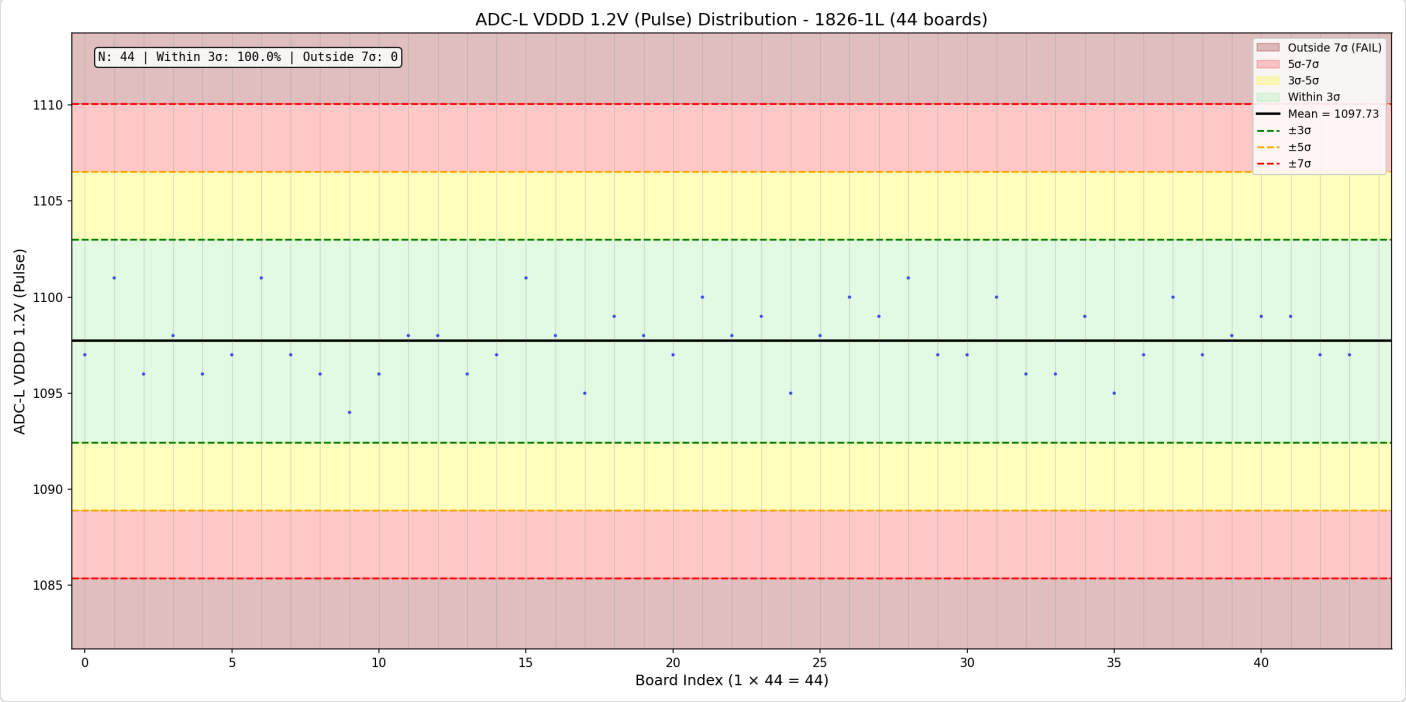
Description: Left ColdADC digital 1.2V supply.

Statistic	Value
Mean	1097.7273
Std	1.7628
Median	1097.5000
Min	1094.0000
Max	1101.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1085.3877	1110.0669

Interpretation: Should be stable within 7σ bounds.



3.5 FERVDDP_pulse (FERVDDP_pulse)

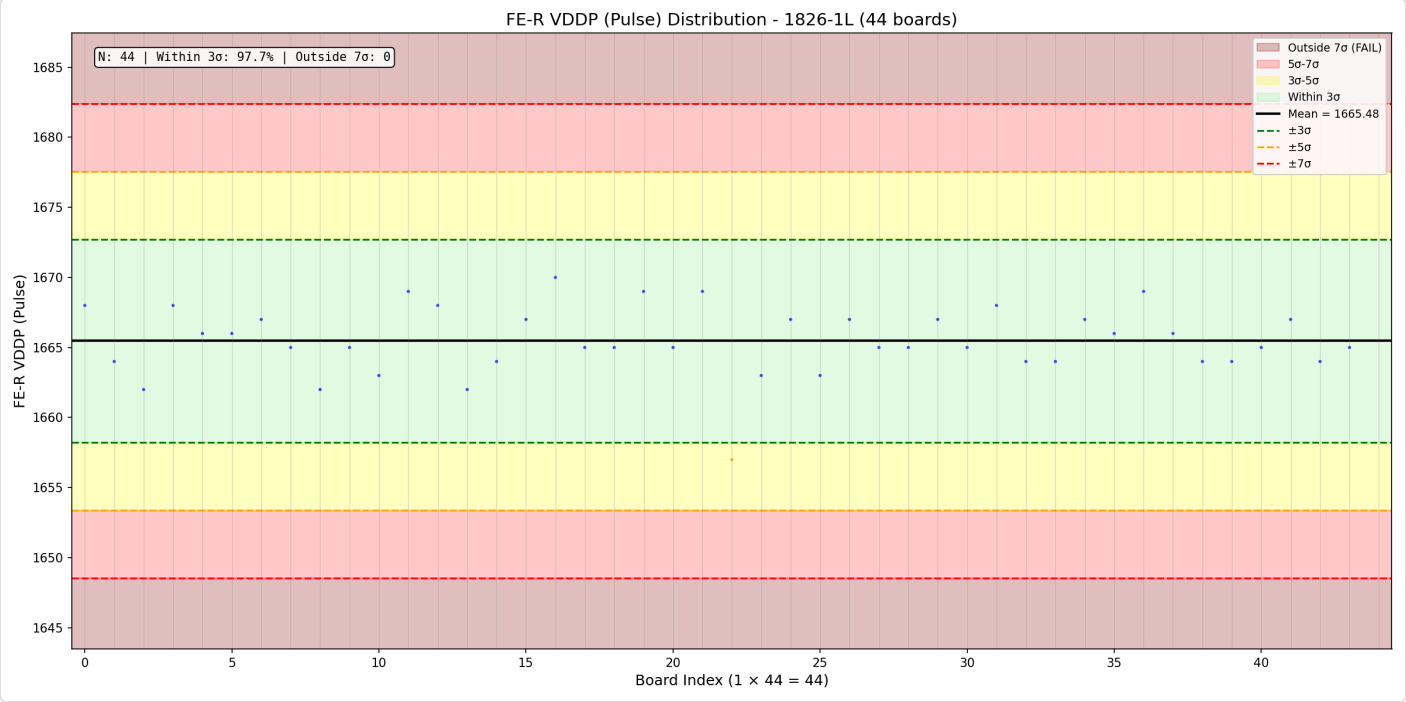
Description: Right LArASIC positive supply voltage.

Statistic	Value
Mean	1665.4773
Std	2.4167
Median	1665.0000
Min	1657.0000
Max	1670.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1648.5604	1682.3942

Interpretation: Higher during pedestal, lower during pulse.



3.6 FELVDDP_pulse (FELVDDP_pulse)

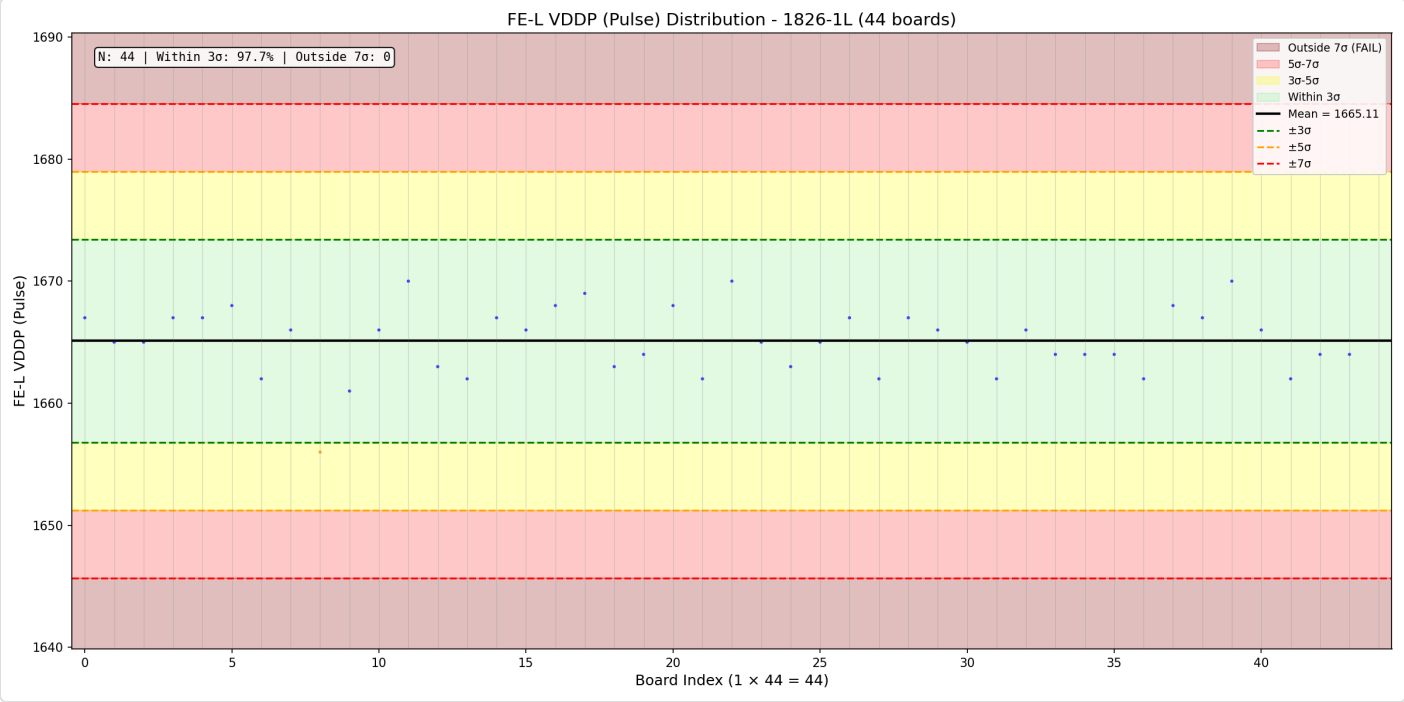
Description: Left LArASIC positive supply voltage.

Statistic	Value
Mean	1665.1136
Std	2.7734
Median	1665.0000
Min	1656.0000
Max	1670.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1645.6998	1684.5274

Interpretation: Higher during pedestal, lower during pulse.



3.7 ADCRP25V_1_HALF_pulse (ADCRP25V_1_HALF_pulse)

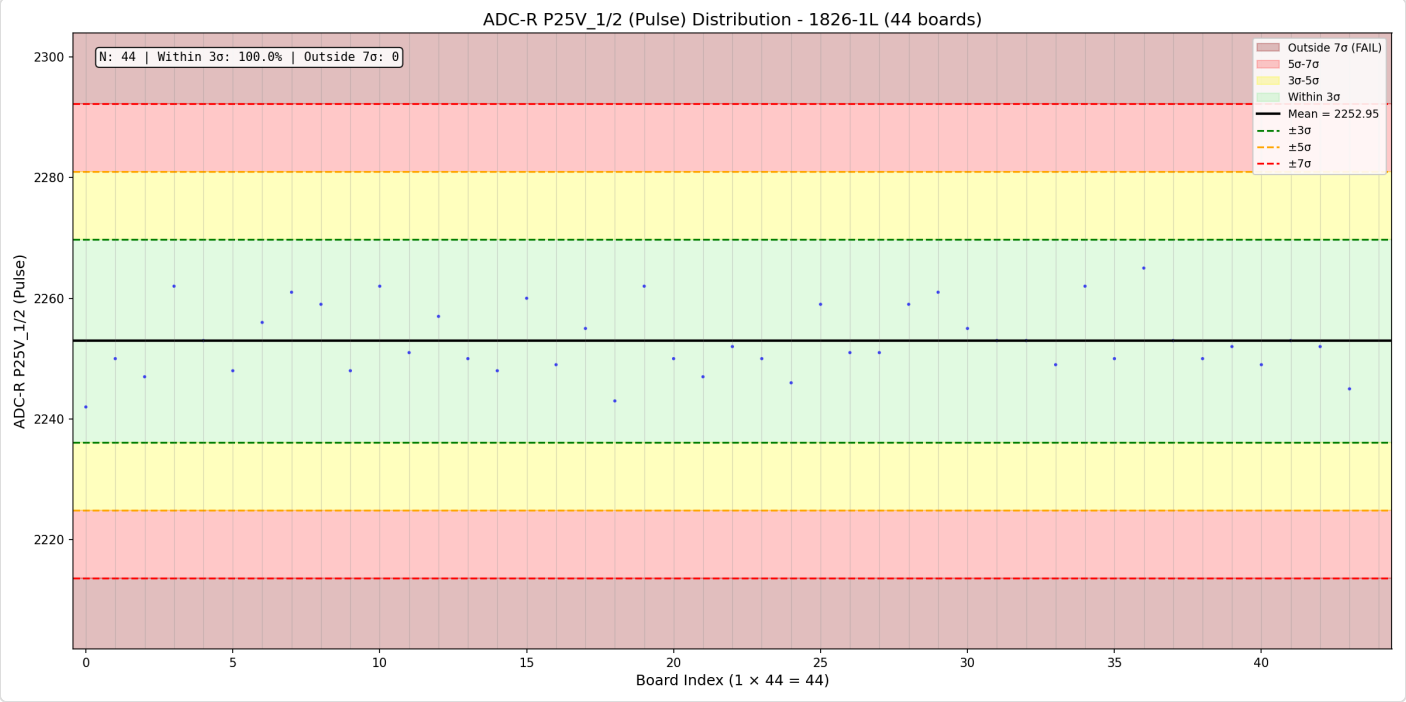
Description: Right ColdADC 2.5V supply #1 (divided by 2).

Statistic	Value
Mean	2252.9545
Std	5.6083
Median	2252.0000
Min	2242.0000
Max	2265.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2213.6964	2292.2126

Interpretation: Should be stable within 7σ bounds.



3.8 ADCLP25V_1_HALF_pulse (ADCLP25V_1_HALF_pulse)

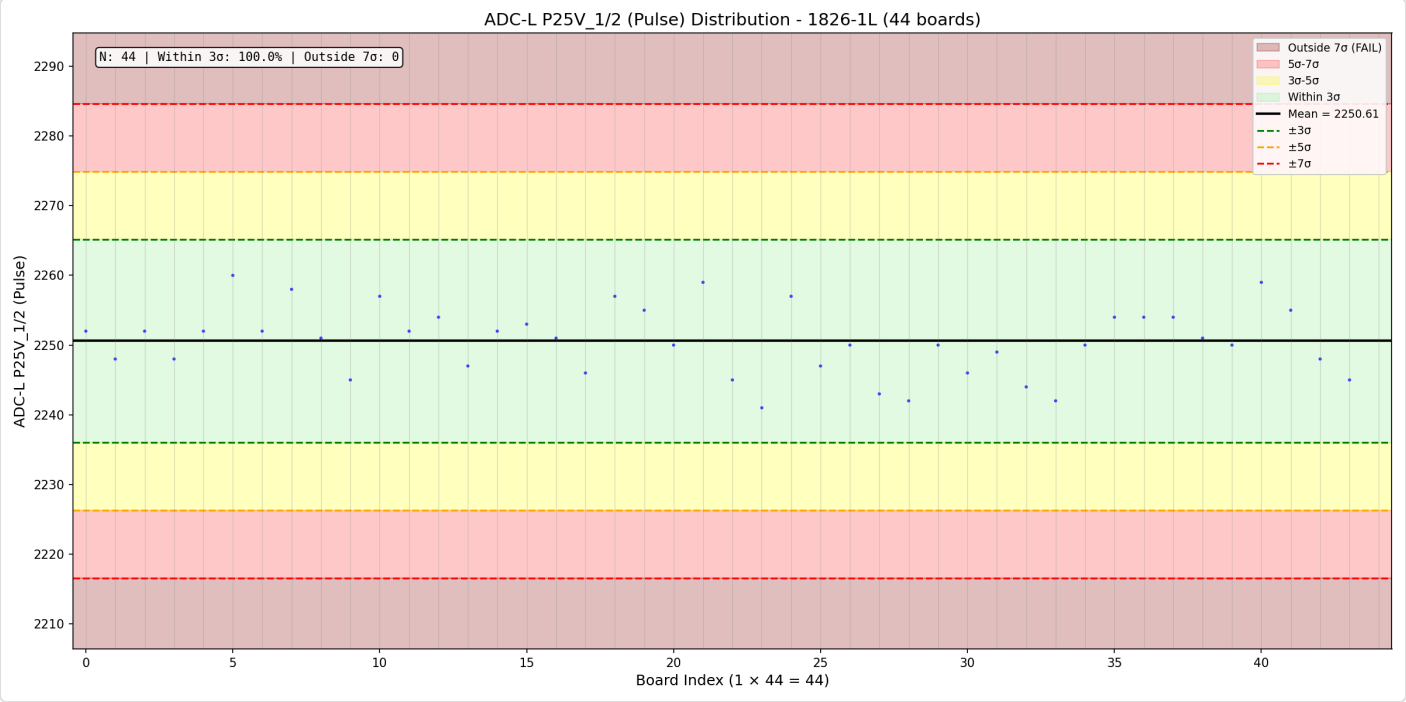
Description: Left ColdADC 2.5V supply #1 (divided by 2).

Statistic	Value
Mean	2250.6136
Std	4.8581
Median	2251.0000
Min	2241.0000
Max	2260.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2216.6069	2284.6203

Interpretation: Should be stable within 7σ bounds.



3.9 ADCRP25V_HALF_pulse (ADCRP25V_HALF_pulse)

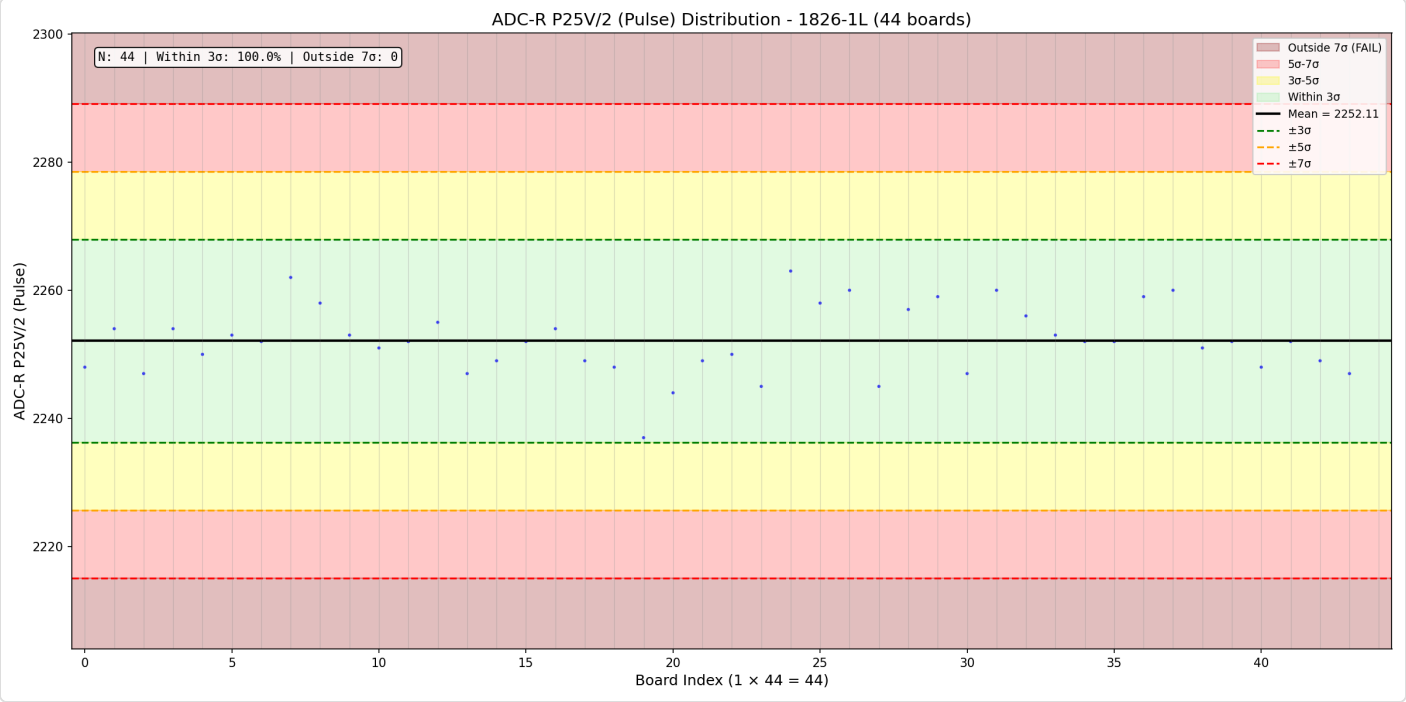
Description: Right ColdADC 2.5V main supply (divided by 2).

Statistic	Value
Mean	2252.1136
Std	5.2838
Median	2252.0000
Min	2237.0000
Max	2263.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2215.1270	2289.1002

Interpretation: Should be stable within 7σ bounds.



3.10 ADCLP25V_HALF_pulse (ADCLP25V_HALF_pulse)

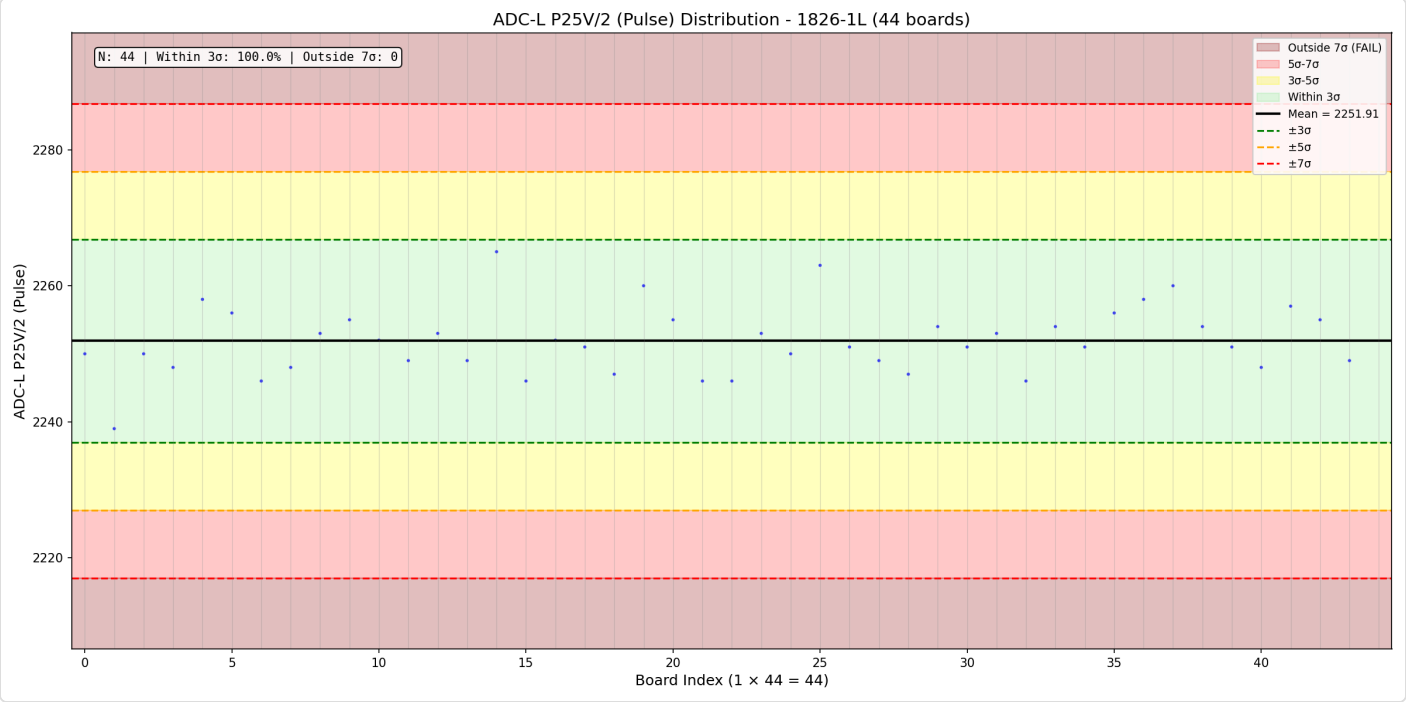
Description: Left ColdADC 2.5V main supply (divided by 2).

Statistic	Value
Mean	2251.9091
Std	4.9764
Median	2251.0000
Min	2239.0000
Max	2265.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
2217.0743	2286.7439

Interpretation: Should be stable within 7σ bounds.



3.11 GND_pulse (GND_pulse)

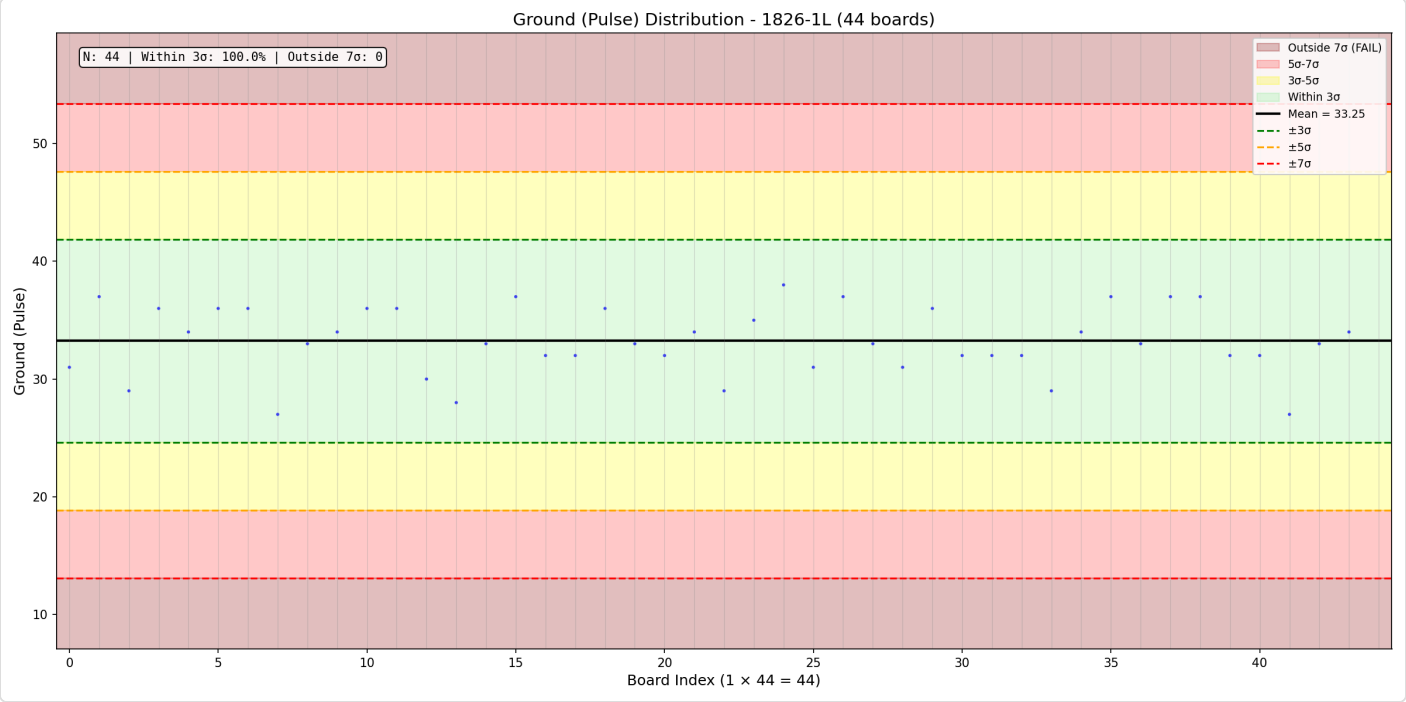
Description: Ground reference measurement.

Statistic	Value
Mean	33.2500
Std	2.8772
Median	33.0000
Min	27.0000
Max	38.0000
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
13.1096	53.3904

Interpretation: Should be near zero, within 7σ bounds.



4. Current Measurements (7σ Tolerance)

Board-level current consumption for each power rail.

4.1 I LArASIC (Ped) (I_LArASIC_ped)

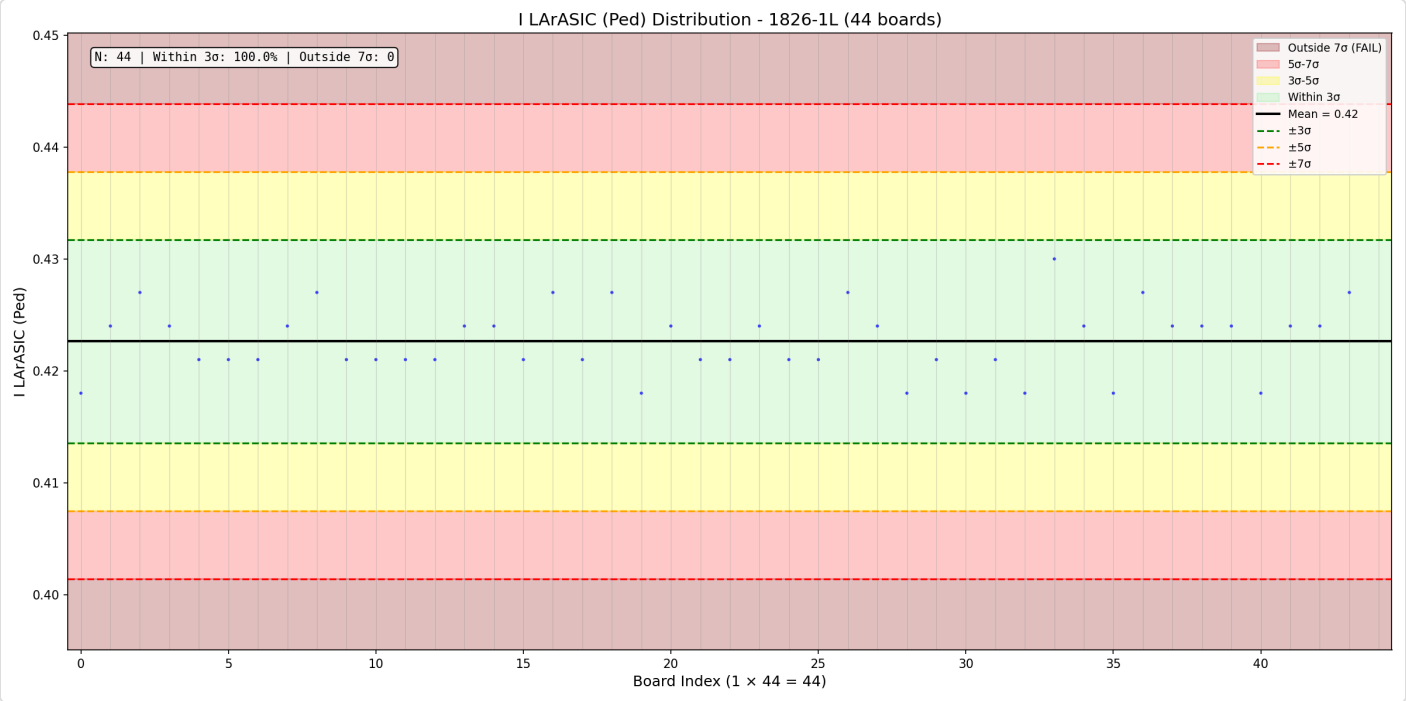
Description: LArASIC front-end ASIC current consumption.

Statistic	Value
Mean	0.4226
Std	0.0030
Median	0.4225
Min	0.4180
Max	0.4300
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
0.4016	0.4436

Interpretation: Should be within 7σ bounds.



4.2 I ColdDATA (Ped) (I_ColdDATA_ped)

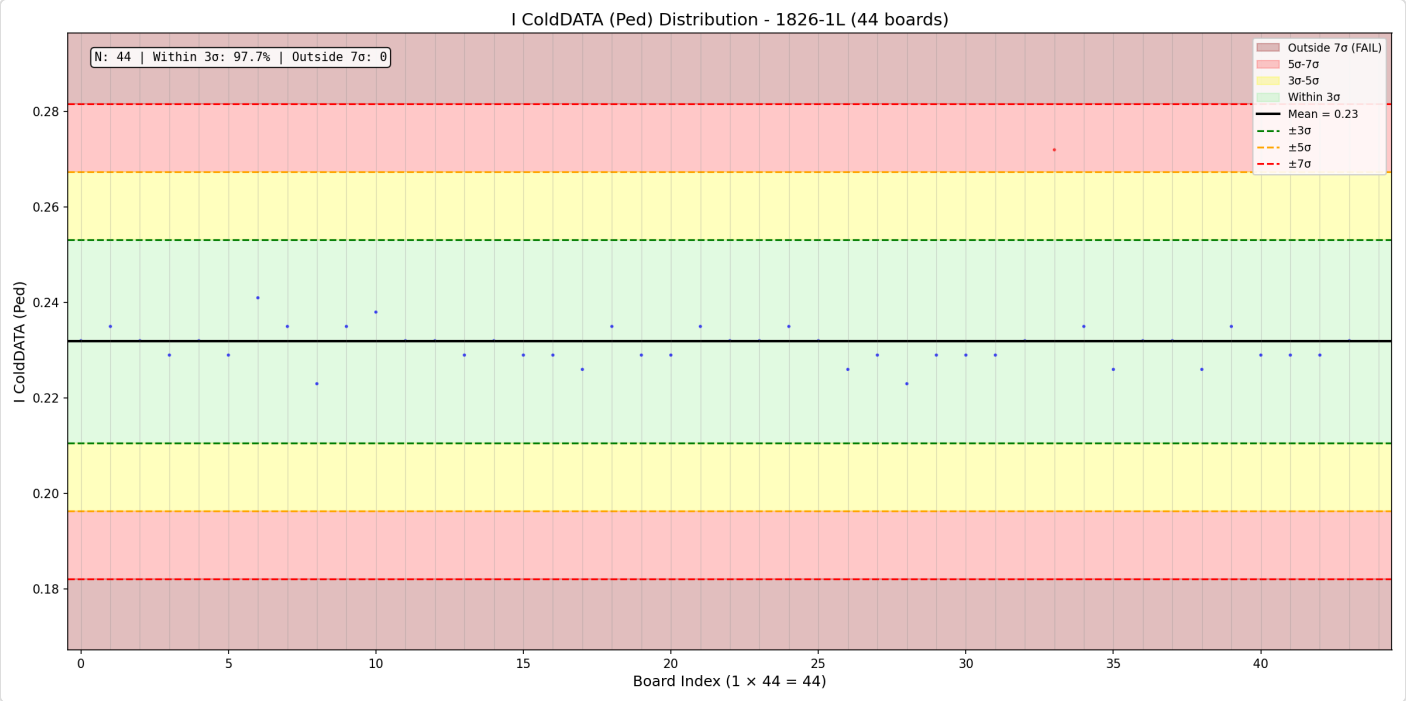
Description: COLDATA serializer ASIC current consumption.

Statistic	Value
Mean	0.2319
Std	0.0071
Median	0.2320
Min	0.2230
Max	0.2720
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
0.1822	0.2816

Interpretation: Should be within 7σ bounds.



4.3 I ColdADC (Ped) (I_ColdADC_ped)

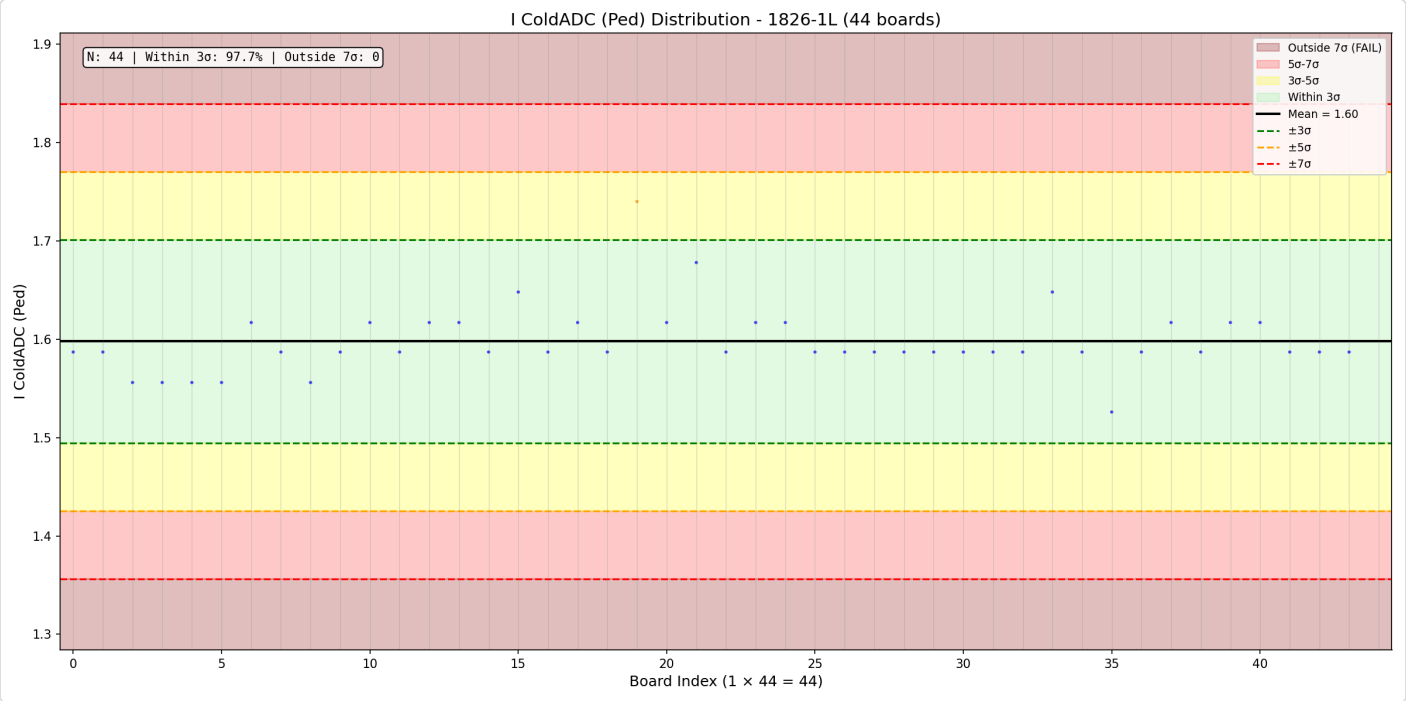
Description: ColdADC digitizer current consumption.

Statistic	Value
Mean	1.5979
Std	0.0345
Median	1.5870
Min	1.5260
Max	1.7400
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1.3564	1.8394

Interpretation: Should be within 7σ bounds.



4.4 I LArASIC (Pulse) (I_LArASIC_pulse)

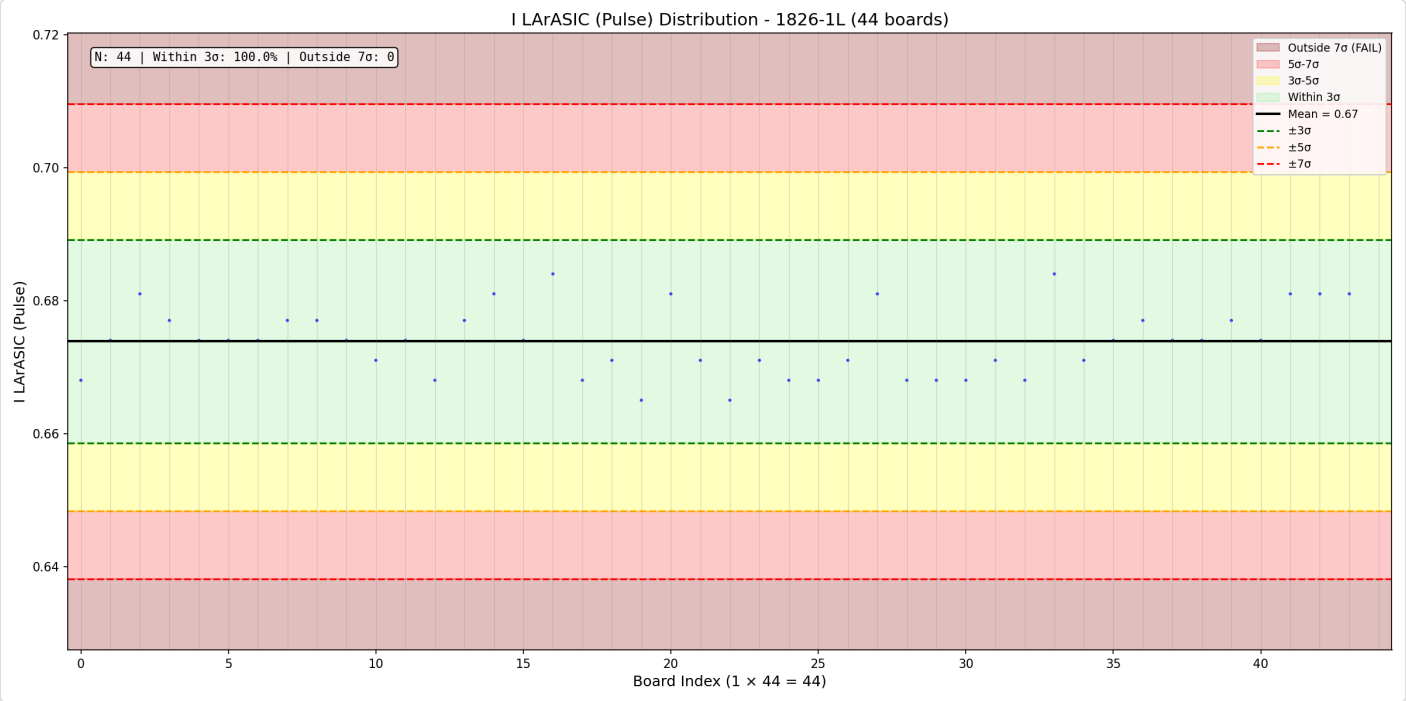
Description: LArASIC front-end ASIC current consumption.

Statistic	Value
Mean	0.6739
Std	0.0051
Median	0.6740
Min	0.6650
Max	0.6840
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
0.6382	0.7096

Interpretation: Should be within 7σ bounds.



4.5 I ColdDATA (Pulse) (I_ColdDATA_pulse)

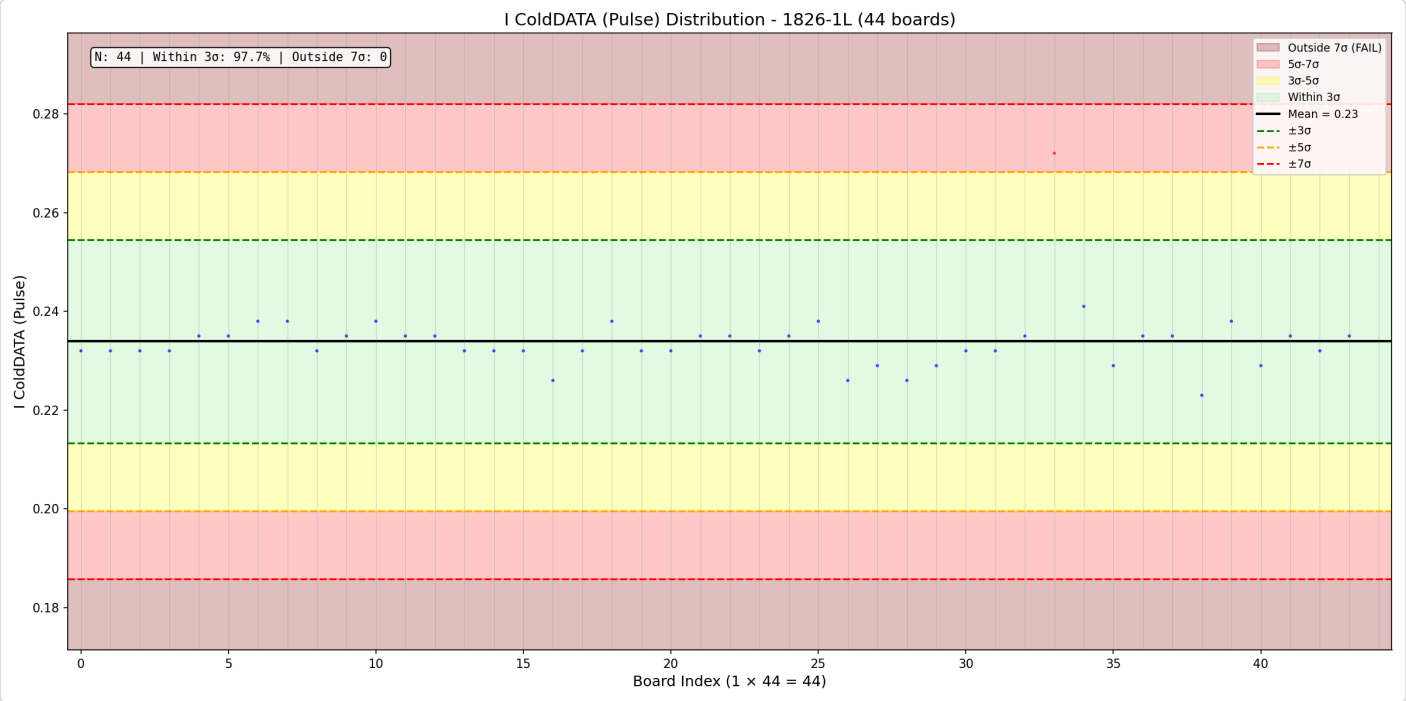
Description: COLDATA serializer ASIC current consumption.

Statistic	Value
Mean	0.2339
Std	0.0069
Median	0.2320
Min	0.2230
Max	0.2720
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
0.1856	0.2822

Interpretation: Should be within 7σ bounds.



4.6 I ColdADC (Pulse) (I_ColdADC_pulse)

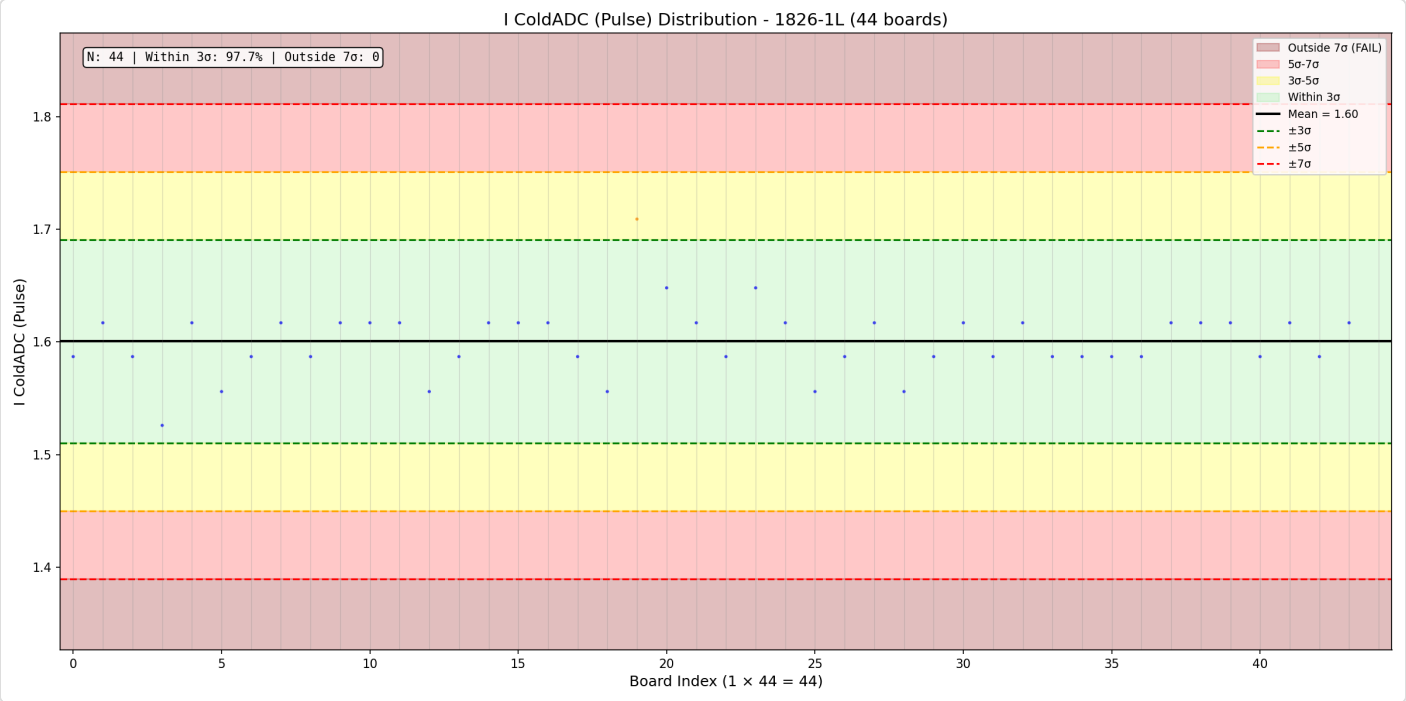
Description: ColdADC digitizer current consumption.

Statistic	Value
Mean	1.6006
Std	0.0301
Median	1.6020
Min	1.5260
Max	1.7090
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
1.3899	1.8113

Interpretation: Should be within 7σ bounds.



5. Power Measurements (7σ Tolerance)

Total board power consumption.

5.1 Total Power (Ped) (Total_Power_ped)

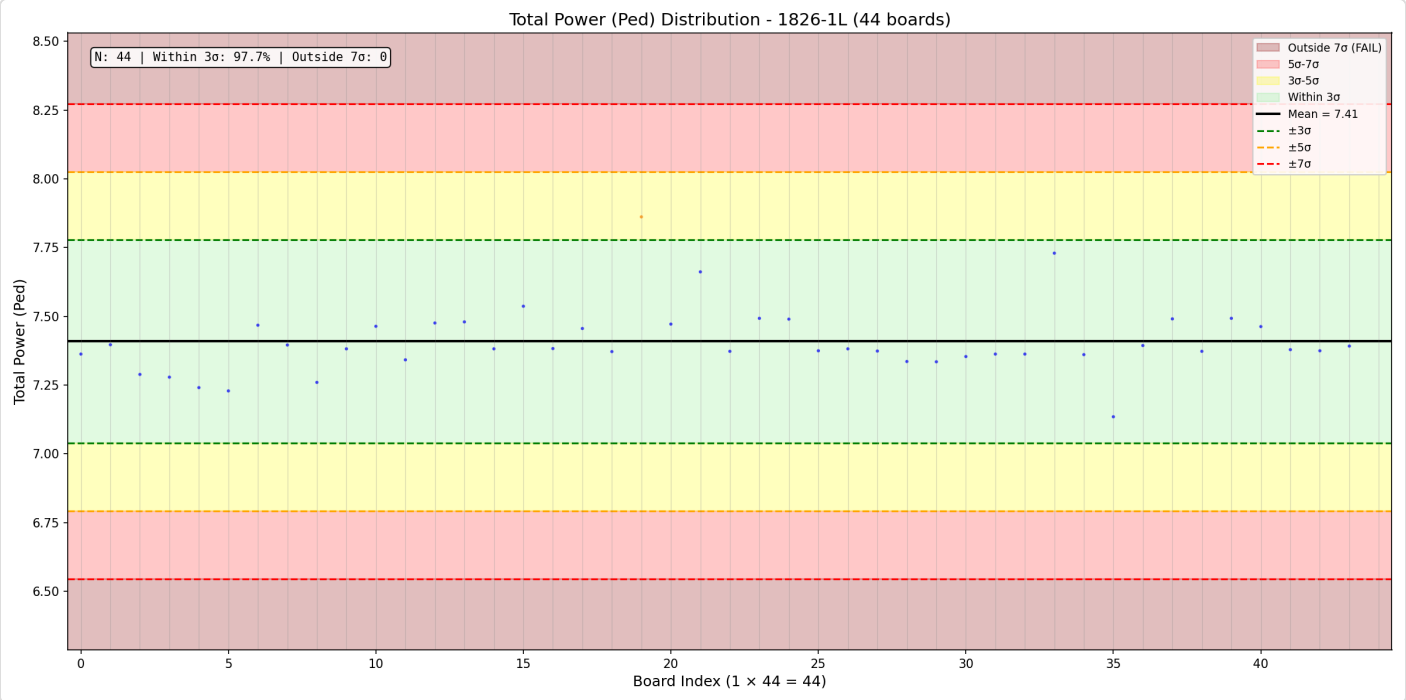
Description: Total board power consumption.

Statistic	Value
Mean	7.4095
Std	0.1233
Median	7.3820
Min	7.1350
Max	7.8620
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
6.5464	8.2726

Interpretation: Should be within 7σ bounds.



5.2 Total Power (Pulse) (Total_Power_pulse)

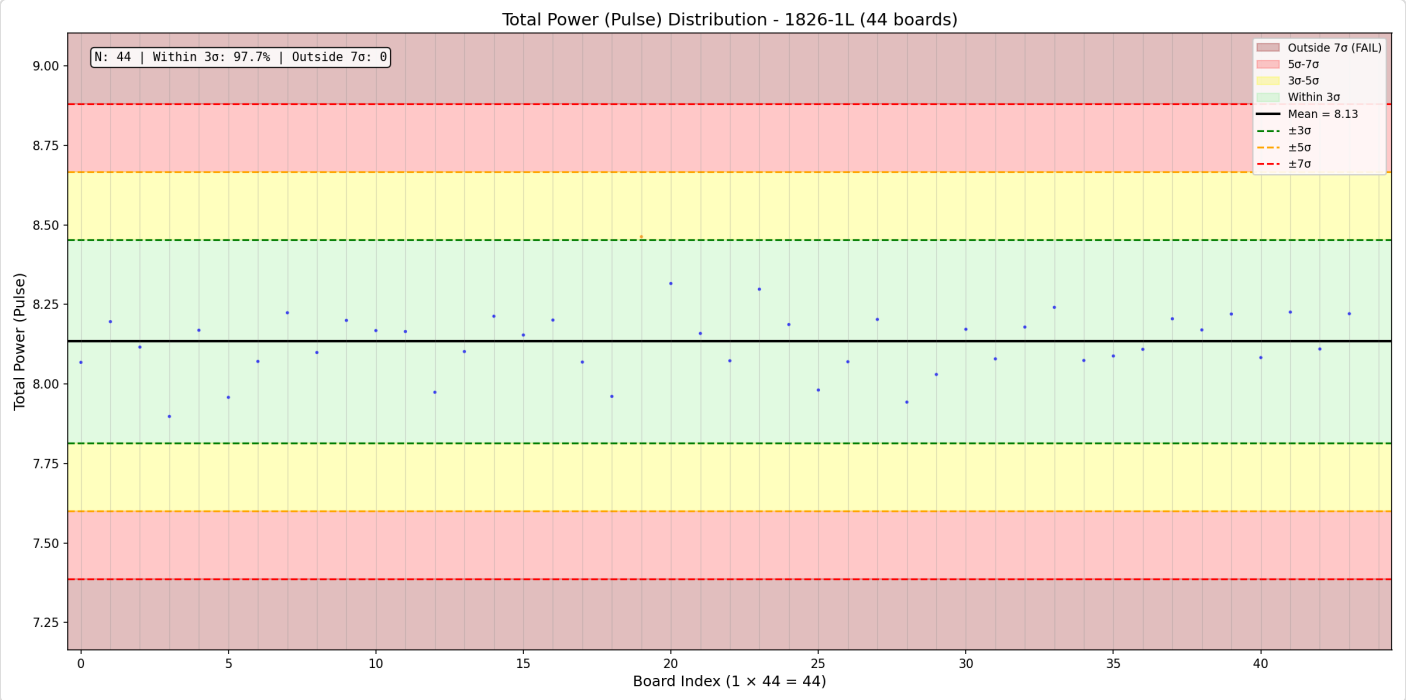
Description: Total board power consumption.

Statistic	Value
Mean	8.1342
Std	0.1066
Median	8.1565
Min	7.8980
Max	8.4630
N	44

Pass/Fail Criteria (7σ):

Lower Limit	Upper Limit
7.3880	8.8804

Interpretation: Should be within 7σ bounds.



6. ASIC-Level Parameters (5σ Tolerance)

Per-ASIC measurements (8 chips per board, 352 total data points).

6.1 FE Temperature (FE T)

Description: LArASIC temperature sensor reading.

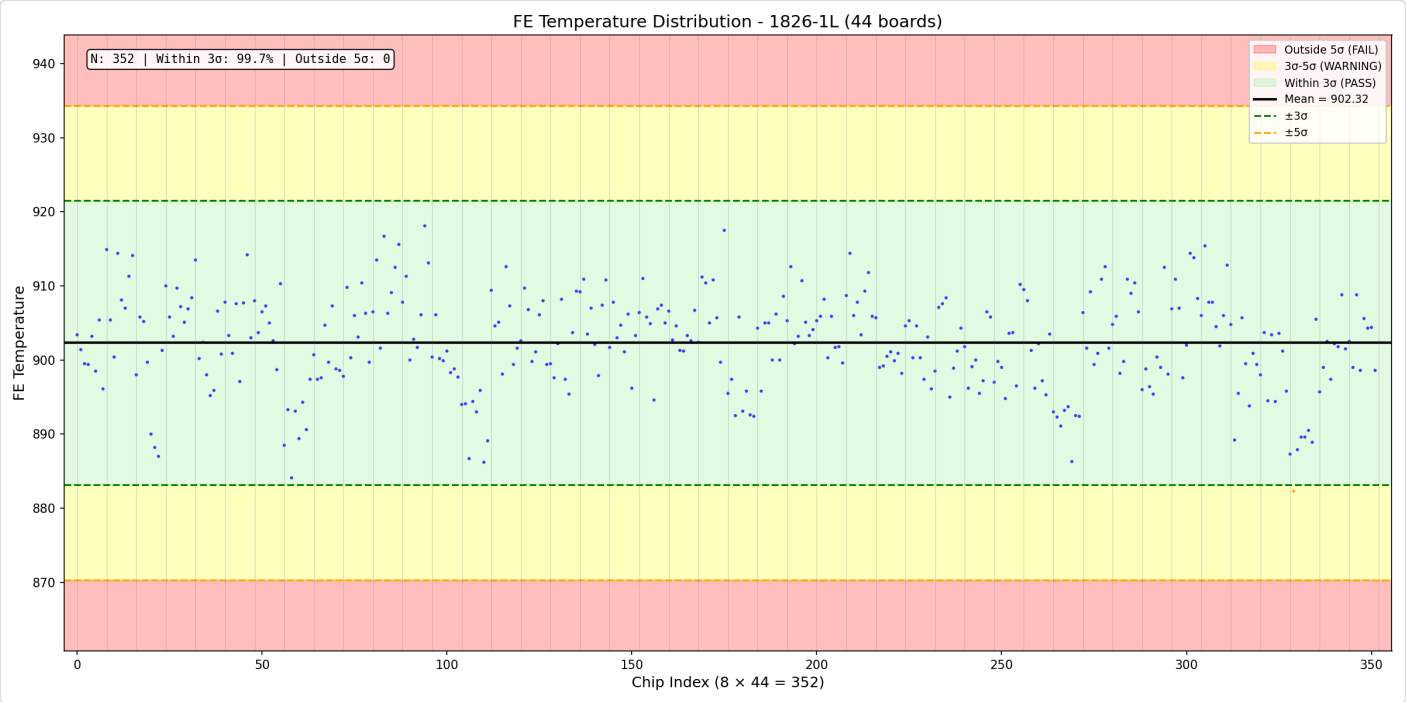
Statistic	Value
Mean	902.3207
Std	6.3980
Median	902.5500
Min	882.3000

Statistic	Value
Max	918.1000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
870.3307	934.3107

Interpretation: Should be consistent at room temperature.



6.2 FE Bandgap (FE BGP)

Description: LArASIC bandgap reference voltage.

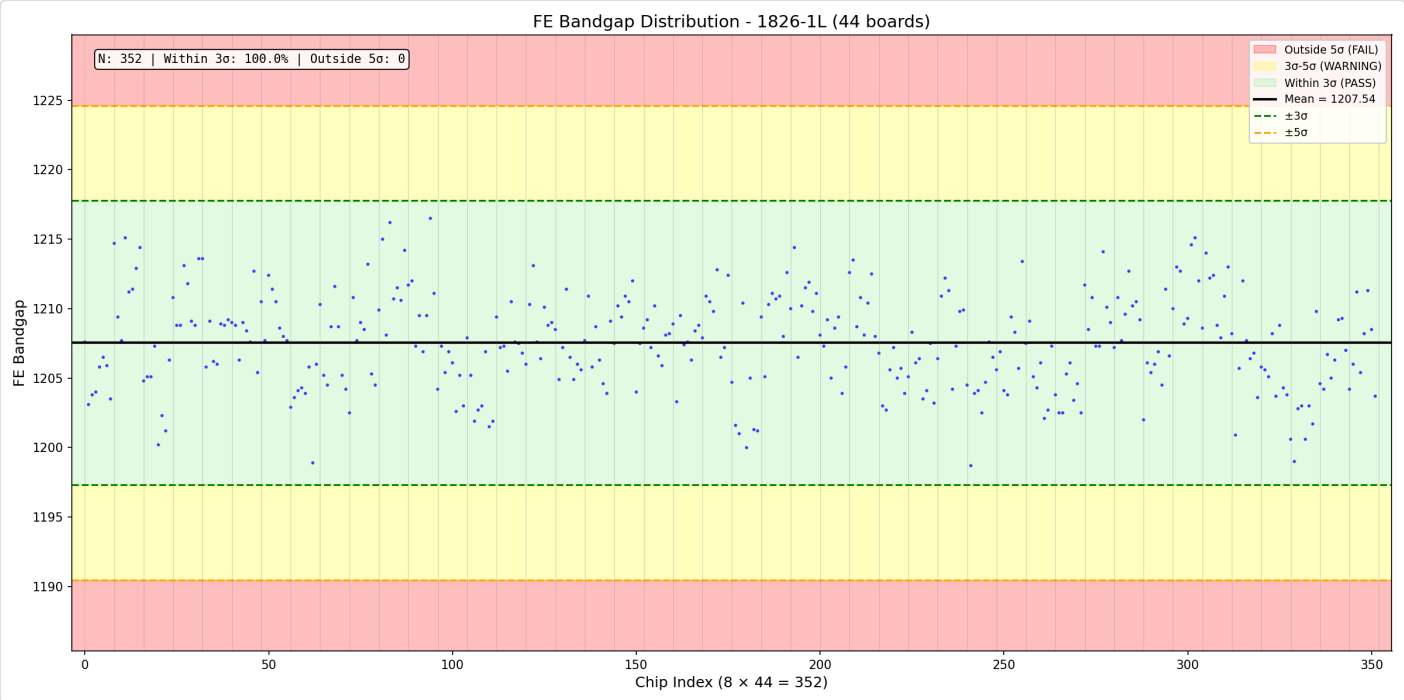
Statistic	Value
Mean	1207.5426
Std	3.4129
Median	1207.5500

Statistic	Value
Min	1198.7000
Max	1216.5000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
1190.4781	1224.6071

Interpretation: Critical for stable operation.



6.3 ADC VCMI (ADC_VCMI)

Description: ColdADC common-mode input voltage.

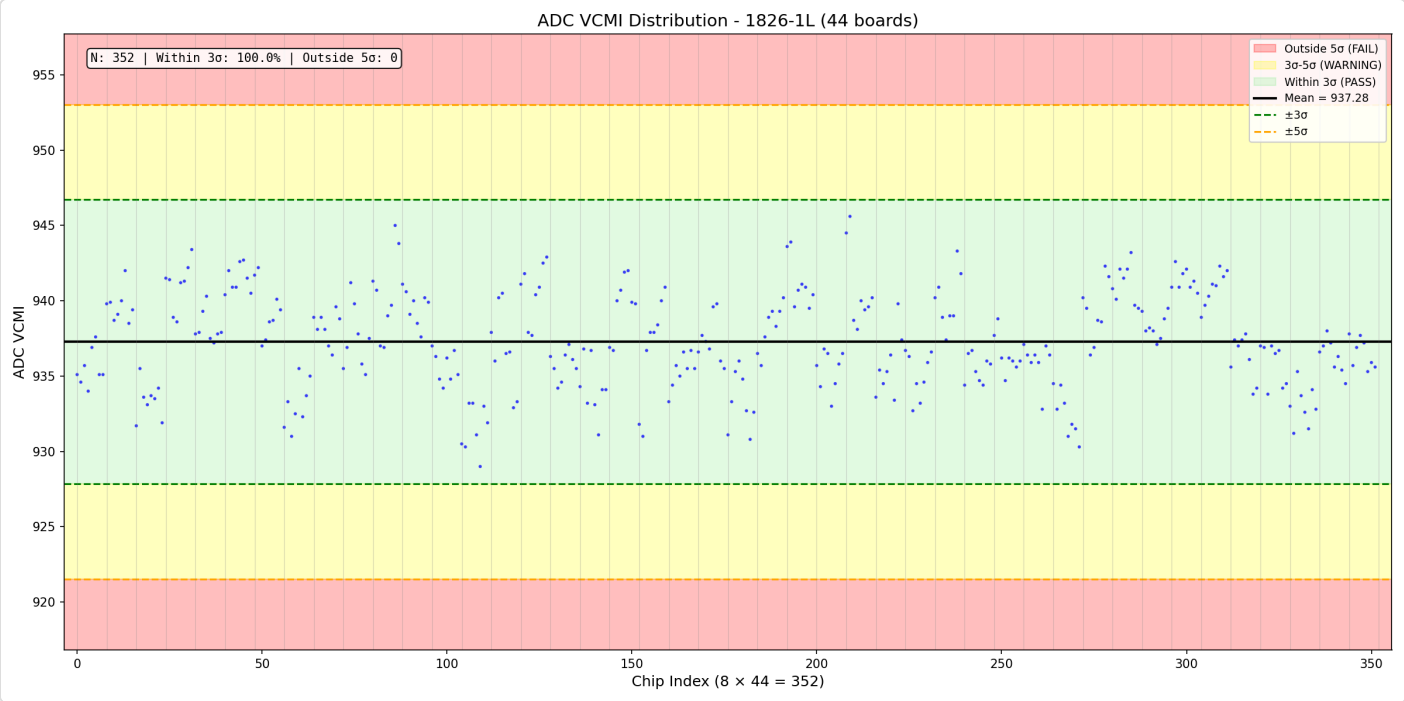
Statistic	Value
Mean	937.2756
Std	3.1470

Statistic	Value
Median	937.0000
Min	929.0000
Max	945.6000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
921.5406	953.0106

Interpretation: Should be consistent across chips.



6.4 ADC VCMO (ADC_VCMO)

Description: ColdADC common-mode output voltage.

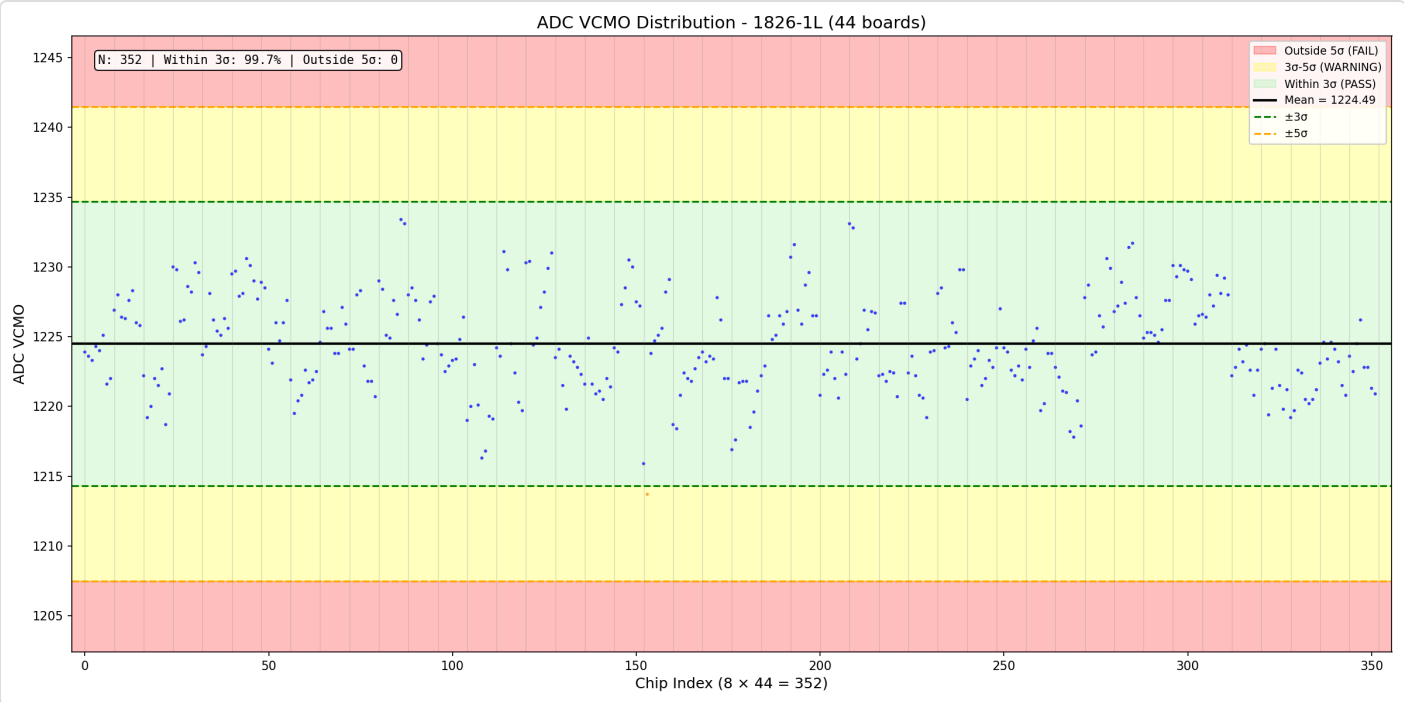
Statistic	Value
Mean	1224.4903

Statistic	Value
Std	3.3991
Median	1224.1000
Min	1213.7000
Max	1233.4000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
1207.4948	1241.4858

Interpretation: Should be consistent across chips.



6.5 ADC VREFP (ADC_VREFP)

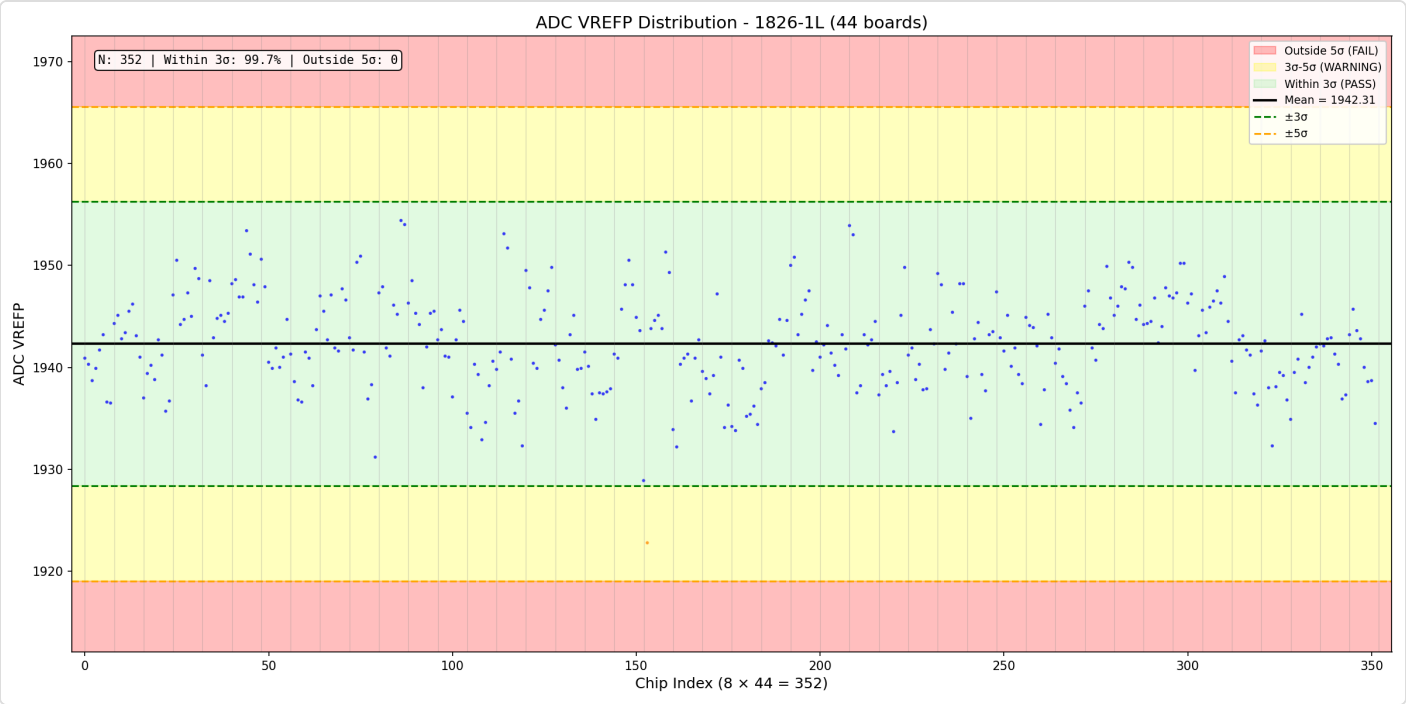
Description: ColdADC positive reference voltage.

Statistic	Value
Mean	1942.3085
Std	4.6486
Median	1942.1500
Min	1922.8000
Max	1954.4000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
1919.0655	1965.5515

Interpretation: Sets upper ADC range limit.



6.6 ADC VREFN (ADC_VREFN)

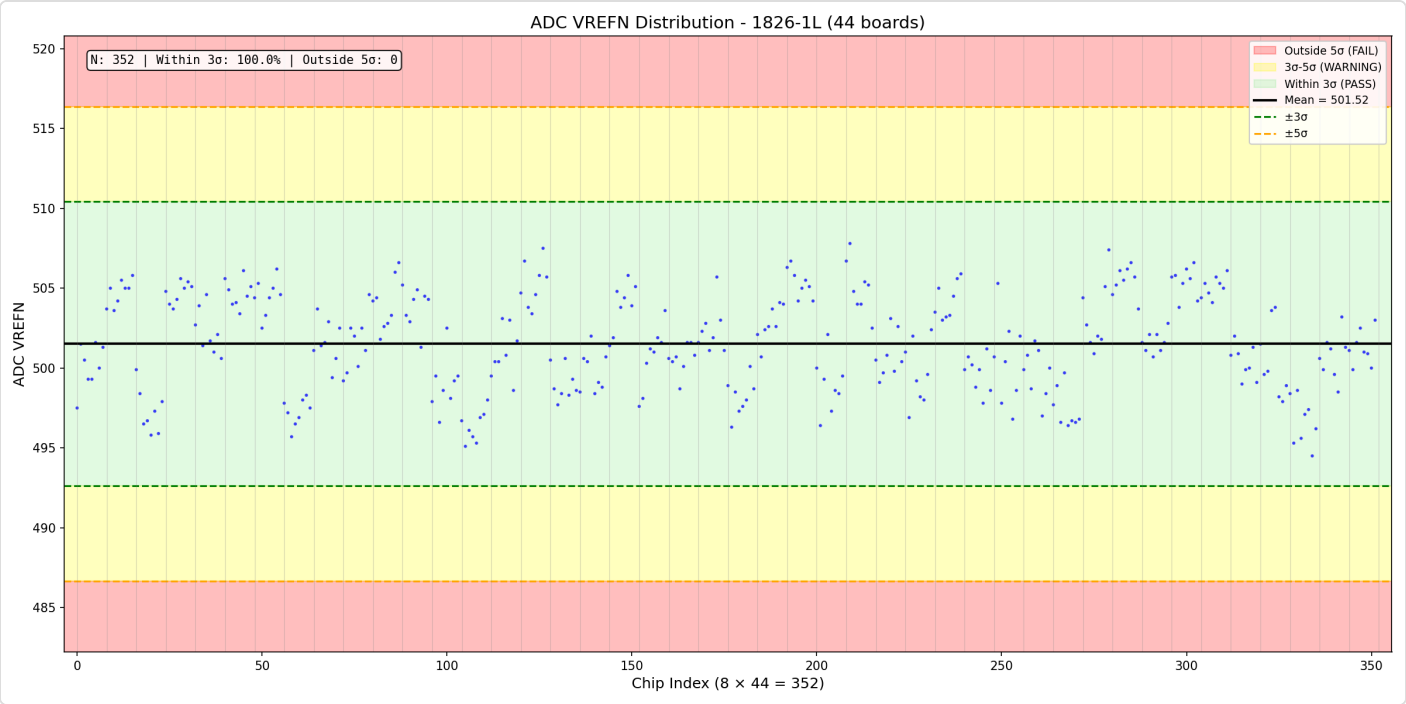
Description: ColdADC negative reference voltage.

Statistic	Value
Mean	501.5187
Std	2.9686
Median	501.5500
Min	494.5000
Max	507.8000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
486.6757	516.3617

Interpretation: Sets lower ADC range limit.



6.7 VSSA (VSSA)

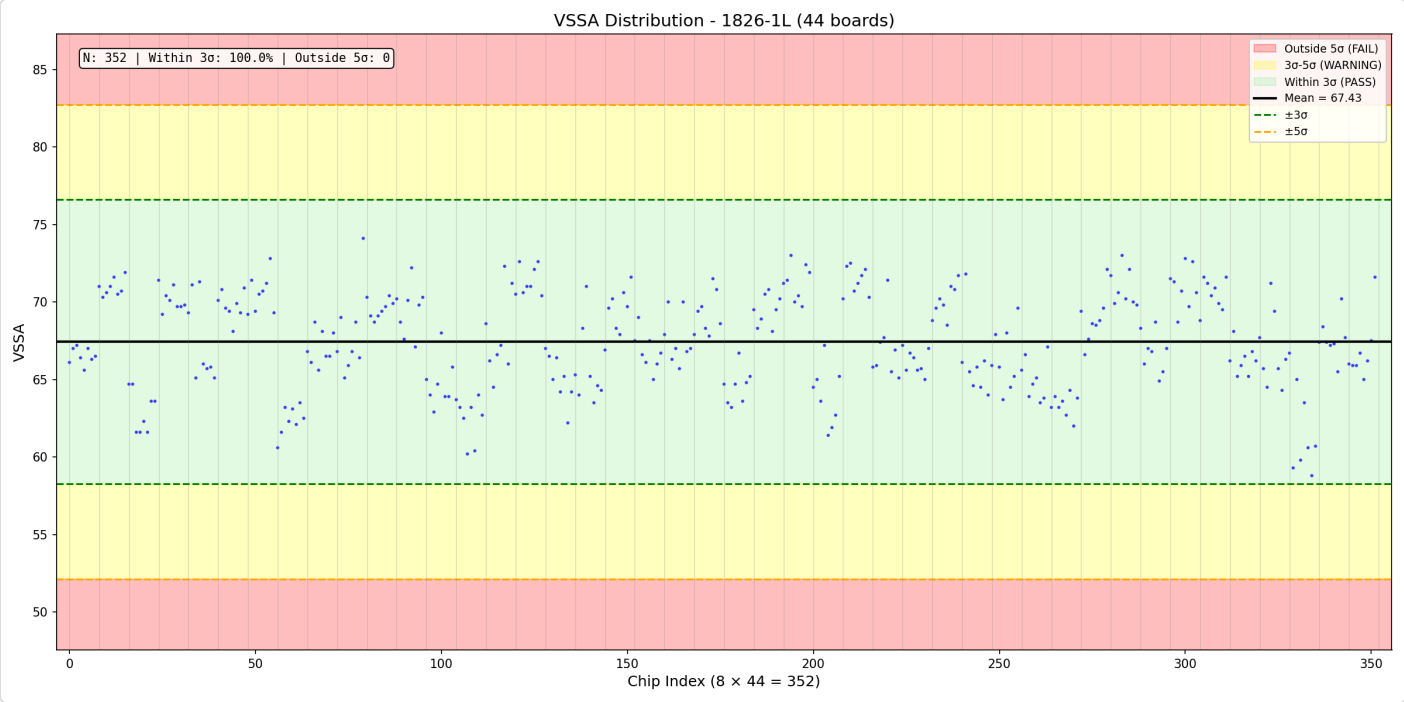
Description: Analog ground reference voltage.

Statistic	Value
Mean	67.4304
Std	3.0570
Median	67.2000
Min	58.8000
Max	74.1000
N	352

Pass/Fail Criteria (5σ):

Lower Limit	Upper Limit
52.1454	82.7154

Interpretation: Should be near zero.



Appendix A: Complete Pass/Fail Criteria Summary

Parameter	Type	Method	Lower Limit	Upper Limit
ped	128ch	5σ	387.8101	1435.8031
rms	128ch	5σ	8.9255	13.4675
PPK_mean	128ch	5σ	12276.2331	13219.0811
NPK_mean	128ch	5σ	3983.4115	4839.6375
BL_mean	128ch	5σ	8229.0016	8930.4476
pedmax	128ch	5σ	426.1951	1472.7801
pedmin	128ch	5σ	348.7288	1399.5968
CDVDDA_ped	ldo_ped	7σ	1171.4041	1198.9141
CDVDDIO_HALF_ped	ldo_ped	7σ	2190.2130	2272.7416
ADCRVDDD1P2_ped	ldo_ped	7σ	1083.9507	1111.3221
ADCLVDDD1P2_ped	ldo_ped	7σ	1084.4961	1110.5949
FERVDDP_ped	ldo_ped	7σ	1772.6051	1814.2131
FELVDDP_ped	ldo_ped	7σ	1769.9499	1815.5955
ADCRP25V_1_HALF_ped	ldo_ped	7σ	2212.7907	2292.3457
ADCLP25V_1_HALF_ped	ldo_ped	7σ	2215.1279	2285.9175
ADCRP25V_HALF_ped	ldo_ped	7σ	2212.4421	2291.2397
ADCLP25V_HALF_ped	ldo_ped	7σ	2215.2669	2289.1421
GND_ped	ldo_ped	7σ	10.8279	48.0357
CDVDDA_pulse	ldo_pulse	7σ	1172.1538	1198.1644
CDVDDIO_HALF_pulse	ldo_pulse	7σ	2194.2329	2269.8581
ADCRVDDD1P2_pulse	ldo_pulse	7σ	1085.8142	1109.8676
ADCLVDDD1P2_pulse	ldo_pulse	7σ	1085.3877	1110.0669
FERVDDP_pulse	ldo_pulse	7σ	1648.5604	1682.3942

Parameter	Type	Method	Lower Limit	Upper Limit
FELVDDP_pulse	ldo_pulse	7σ	1645.6998	1684.5274
ADCRP25V_1_HALF_pulse	ldo_pulse	7σ	2213.6964	2292.2126
ADCLP25V_1_HALF_pulse	ldo_pulse	7σ	2216.6069	2284.6203
ADCRP25V_HALF_pulse	ldo_pulse	7σ	2215.1270	2289.1002
ADCLP25V_HALF_pulse	ldo_pulse	7σ	2217.0743	2286.7439
GND_pulse	ldo_pulse	7σ	13.1096	53.3904
I_LArASIC_ped	current	7σ	0.4016	0.4436
I_ColdDATA_ped	current	7σ	0.1822	0.2816
I_ColdADC_ped	current	7σ	1.3564	1.8394
I_LArASIC_pulse	current	7σ	0.6382	0.7096
I_ColdDATA_pulse	current	7σ	0.1856	0.2822
I_ColdADC_pulse	current	7σ	1.3899	1.8113
Total_Power_ped	power	7σ	6.5464	8.2726
Total_Power_pulse	power	7σ	7.3880	8.8804
FE T	8chip	5σ	870.3307	934.3107
FE BGP	8chip	5σ	1190.4781	1224.6071
ADC_VCMI	8chip	5σ	921.5406	953.0106
ADC_VCMO	8chip	5σ	1207.4948	1241.4858
ADC_VREFP	8chip	5σ	1919.0655	1965.5515
ADC_VREFN	8chip	5σ	486.6757	516.3617
VSSA	8chip	5σ	52.1454	82.7154

Appendix B: Tolerance Configuration

Parameter Type	Tolerance	Description
128-Channel	5σ	Mean ± 5×Std, for large datasets (5632 points)
8-Chip ASIC	5σ	Mean ± 5×Std, for ASIC-level measurements (352 points)
LDO Voltages	7σ	Mean ± 7×Std, wider tolerance for power rail (44 points)
Currents	7σ	Mean ± 7×Std, wider tolerance for current measurements
Power	7σ	Mean ± 7×Std, wider tolerance for power measurements

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Appendix C: Board List

44 boards were used in this analysis (latest passing test only). See `good_boards_list.txt` for the complete list.

Appendix D: Excluded Boards

- **Board 00046:** Manually excluded (known bad board)
- Boards with only failed tests were automatically excluded

Report generated from QC checkout data analysis BNL Cold Electronics Testing Facility